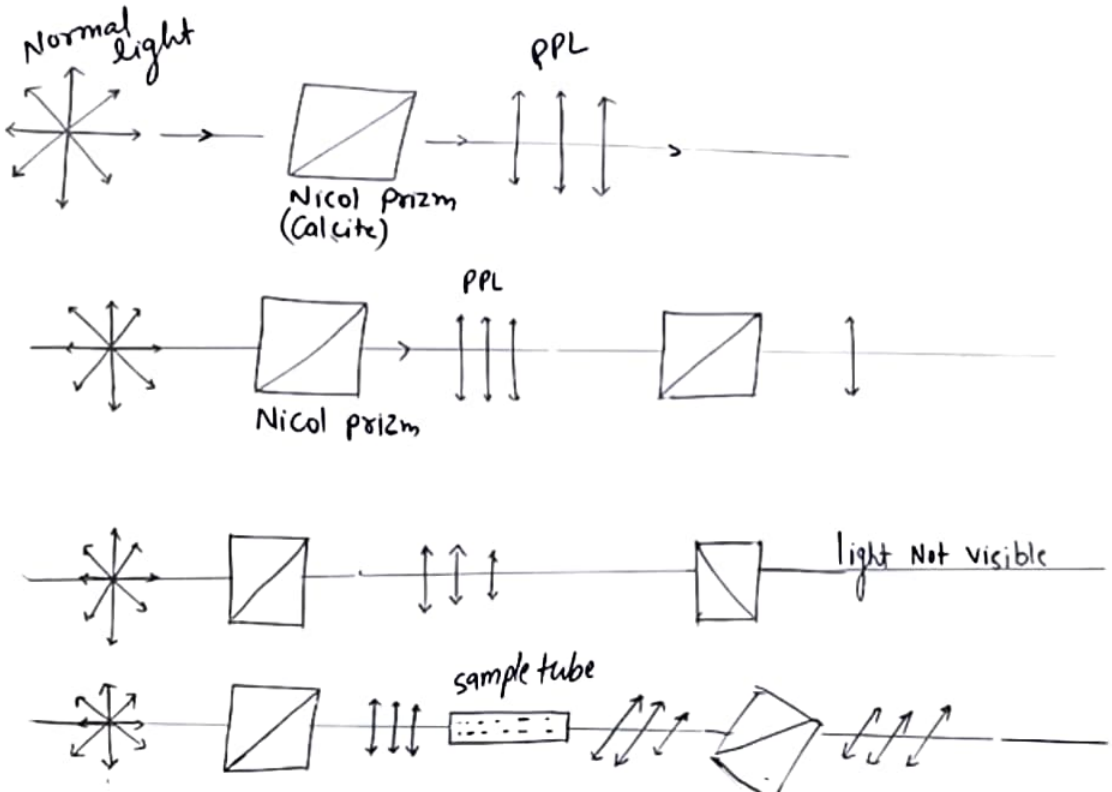
OPTICAL ISOMERISM

(I) Optical activity:- Compounds which can rotate vibrational plane of Plane polarised light (PPL) are called optical active compounds and this phenomenon is called optical activity.

Stereo isomers which have different optical activity are called optical isomers.

(II) Plane Polarised Light (PPL):- Light wave whose vibration occurs only in single plane to perpendicular w.r.t. propagation light

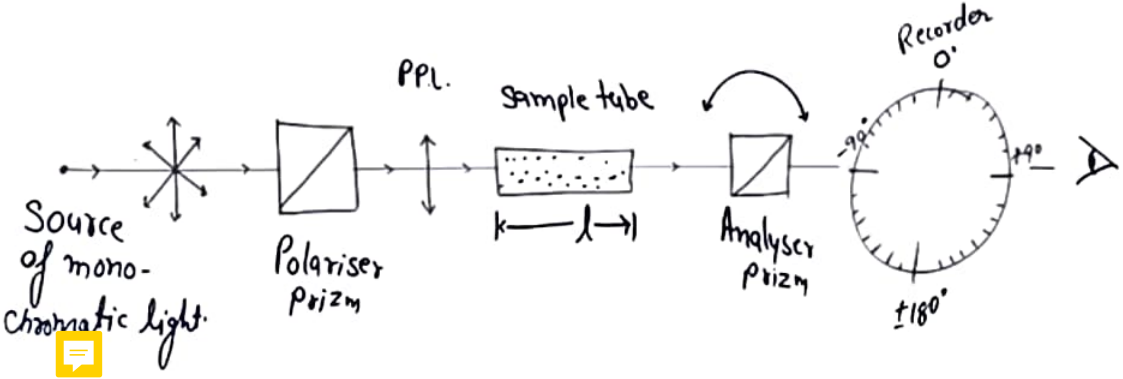
OPTICAL ISOMERISM



OPTICAL ISOMERISM

(III) Polarimeter :-

By using this instrument we can measure the value of observed rotation of vibrational plane of PPL in presence of optical active compound.

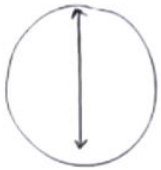


OPTICAL ISOMERISM

(IV) Observation on polarimeter:-

For Optical inactive comp (O.I.C)

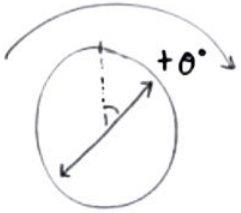
$\theta = 0^\circ$



θ = Obs. rotation

For Optical Active Comp (OAC)

For Clockwise rotation



Dextrorotatory comp.

(d) or (+)

For Anticlockwise rotation



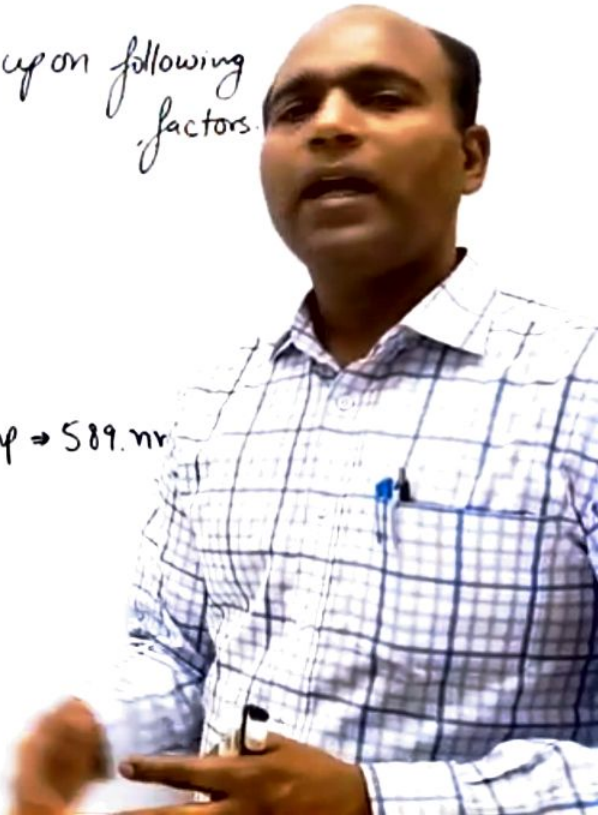
Laevorotatory comp.

l or (-)

OPTICAL ISOMERISM

⇒ The magnitude of optical rotation depend upon following factors.

1. Nature of substance
2. Length of sample tube
3. Concen. of solution.
4. Wave length of light. (usually Na-D lamp ⇒ 589 nm
5. Solvent
6. Temp etc.



OPTICAL ISOMERISM

(I) Specific rotation $[\alpha]$

$$\theta \propto l$$

$$\theta \propto c$$

$$\theta \propto c \cdot l$$

$$\theta = [\alpha]\, c \cdot l$$

$$[\alpha]_D^{t=25^\circ C} = \frac{\theta}{c \cdot l}$$

θ = Obs. rotation (degree)

c = Concentration of sample (in g/ml)

l = length of sample tube (in dm)

1 dm = 10 cm

NEERAJ SAINI (N.S. SIR)

OPTICAL ISOMERISM

(I) Specific rotation [α]

Q.1 For a solution of an optical active compound the value of optical rotation 13.4°. Conc of solution 2g in 10ml and length of sample tube 25 cm. Sepecific rotation = ?

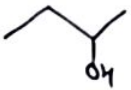
Sol $[\alpha] = \frac{\theta}{c \cdot l} = \frac{13.4^\circ}{\frac{2}{10} \times \frac{25}{10}} = \frac{13.4 \times 100}{2 \times 25} = 26.8^\circ$

Q. Identify the value of specific rotation of a 1M Solution of 2-butanol

length of sample tube = 5 cm

Obs rotation = +7.4°

Sol $[\alpha] = \frac{\theta}{c \cdot l}$



$C_4H_{10}O$

48
10
16
74

$c = 1M = 1 mol/1 lit$

$c = 74 g/1000 ml$

$l = \frac{5}{10}$ dm.

OPTICAL ISOMERISM

(I) Specific rotation $[\alpha]$

$$[\alpha] = \frac{74}{\frac{5}{10} \times \frac{74}{1000}} = \frac{74 \times 1000 \times 2}{74 \times 1} = +200^\circ$$

Q. Identify the value of specific rotation of a 1M solution of 2-butanol

length of sample tube = 5 cm

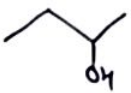
Obs rotation = +7.4°

Sol. $[\alpha] = \frac{\theta}{c \cdot \ell}$

$c = 1M = 1 mol / 1 lit$

$c = 74 g / 1000 ml$

$\ell = \frac{5}{10}$ dm.



$C_4H_{10}O$

48
10
16
74

OPTICAL ISOMERISM

(VI) Chiral carbon = Asymmetric carbon:-

Four different group containing tetrahedral carbon is called chiral carbon.

Note 1 : Only one chiral atom containing compounds are always optically active.

Note 2 : When more than one chiral atoms are present

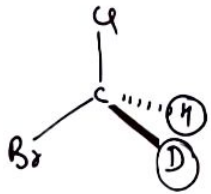
- with dissimilar ends → always optically active
- with similar ends. → may be optically active may no be active (Consider 3D structure)

OPTICAL ISOMERISM

(VI) Chiral carbon = Asymmetric carbon:-

Q Identify no. of chiral carbons.

eg 1



One

O.A.C

eg 2



Two

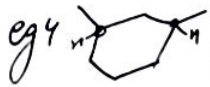
O.A.C

eg 3



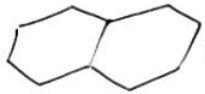
Zero

eg 4



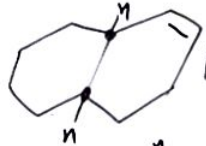
Two

eg 5



zero

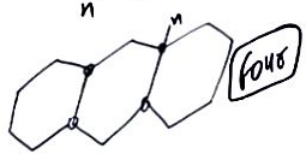
eg 6



Two

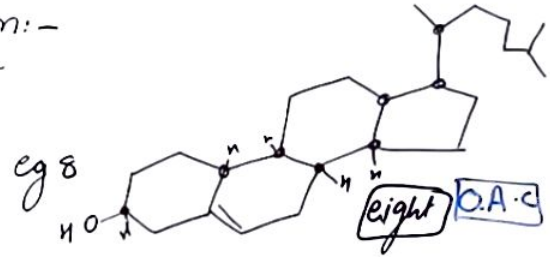
OAC

eg 7



Four

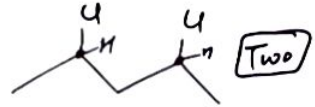
eg 8



eight

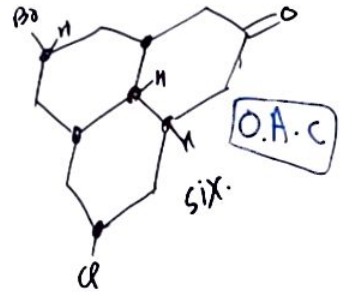
O.A.C

eg 9



Two

eg 10



O.A.C

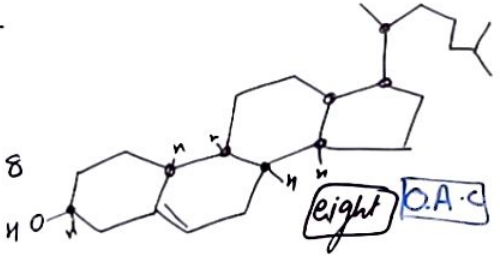
six.

OPTICAL ISOMERISM

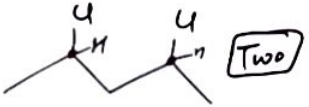
(VI) Chiral carbon = Asymmetric carbon:-

Q Identify no. of Chiral carbons.

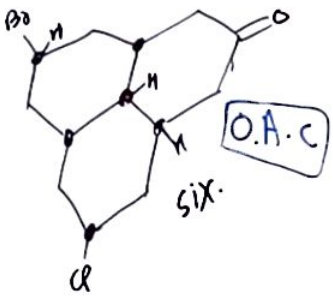
eg 8



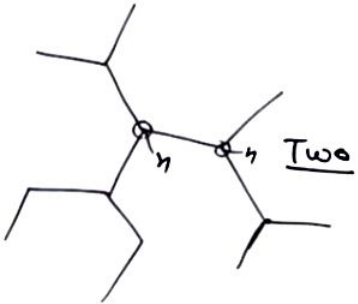
eg 9



eg 10



eg 11

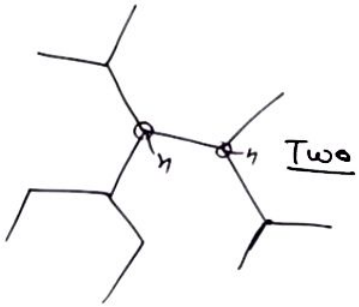


OPTICAL ISOMERISM

(VI) Chiral carbon = Asymmetric carbon:-

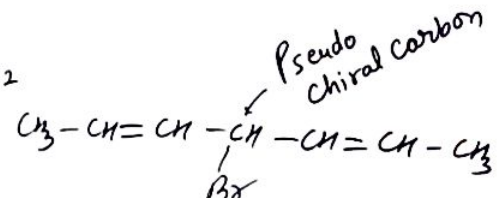
Q Identify no. of chiral carbons.

eg 11



Two

eg 12



① cis

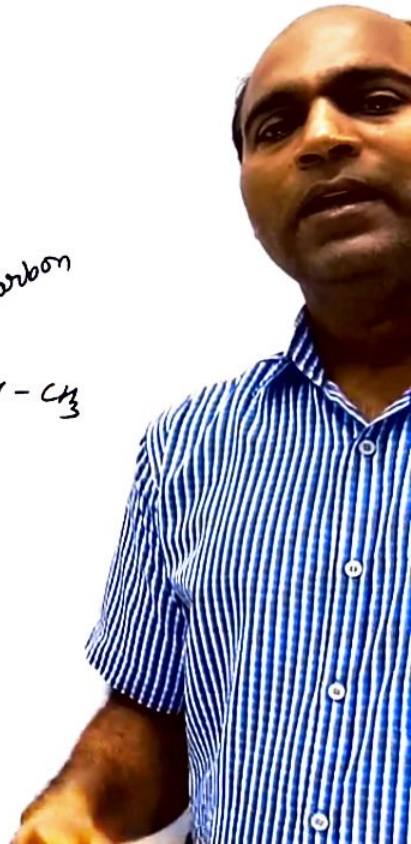
② T

③ ✓ cis

cis

T

Trans



OPTICAL ISOMERISM

(VII) Condition for optical activity :-

- Optical active compound are not superimpose on its mirror image structure. This property is called chirality.
- All chiral molecules are Optical active.

Chirality is necessary and sufficient condition for optical
ity

OPTICAL ISOMERISM

(VII) Condition for optical activity :-

Center of symm., plane of sym. and alternating axis of symmetry are not present in optical active compounds (Chiral molecule.

Out of these elements of symm. if any one element of sym. will present in molecule, then compound will optically inactive

OPTICAL ISOMERISM

(VIII) Elements of symmetry:-

(A) Axis of symmetry (C)

(B) Plane " " (σ)

(C) Center of " (i)

(D) Alternating axis of symmetry (S)

} useful for identification of optical active comp.

OPTICAL ISOMERISM

(VIII) Elements of symmetry:-

(A) Axis of symmetry (C)

It is an imaginary axis which passed through a molecule. When we rotate this molecule around this axis by 360° then we optain identical arrangement (more than one) with respect to original arrangement.

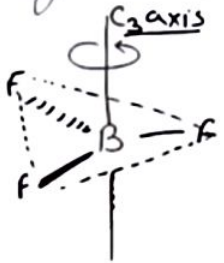
OPTICAL ISOMERISM

(VIII) Elements of symmetry:-

(A) Axis of symmetry (C)

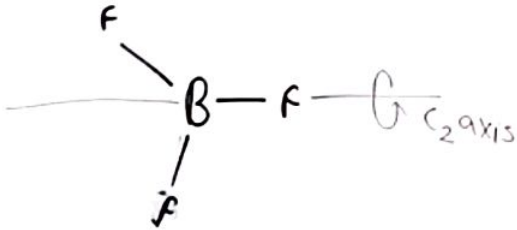
C_n

eg 1 BF_3



→ one C_3 axis

$n = \frac{360}{120}$



$= C_{\frac{360}{180}}$

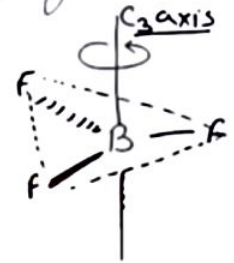
$n = \frac{360}{180}$

OPTICAL ISOMERISM

(VIII) Elements of symmetry:-

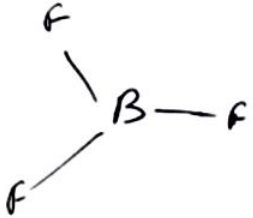
(A) Axis of symmetry (C$_n$)

eg 1 BF_3



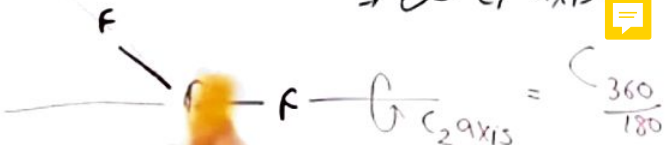
⇒ one C_3 axis

$n = \frac{360}{120}$



⇒ Three C_2 axis

⇒ ∞ C_1 axis



$C_{\frac{360}{180}}$

$n = \frac{360}{180}$

OPTICAL ISOMERISM

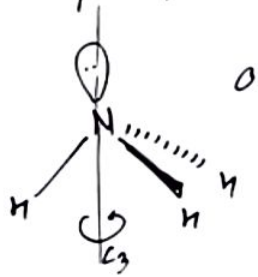
(I) Elements of symmetry:-

eg 2



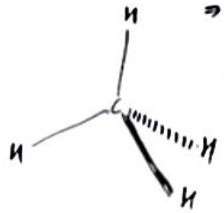
one C_2 axis

eg 3

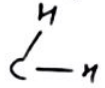


one C_3 axis

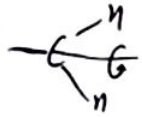
eg 4 CH_4



⇒ Bond angle : 109.5°

⇒  bond angles = 6

⇒ 4 - C_3 axis

⇒ Three C_2 axis 

OPTICAL ISOMERISM

(VIII) Elements of symmetry:-



A ~~B~~ C D E F G

$C_2 = V$ = A B C D E H I K M N O S T U V W X Y Z

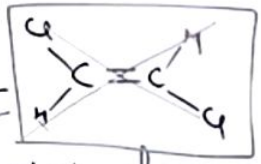
$C_2 = X$ = F G J L P Q R

OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

(B.) **Plane of Symmetry :-** It is an imaginary plane which bisects the molecule/Object in two equal half parts and these both half parts should be mirror image of each other.

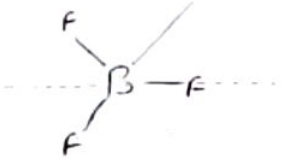
(C) **Centre of symmetry = Point of symmetry = Inversion centre :-**



It is an imaginary point in the centre of molecule from which if line is drown to the direction of a perticular group and it is extended in opposite direction, it meets two mirror image groups, on equal distance from centre of molecule.

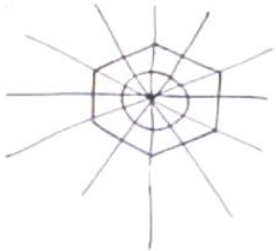
OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

eg 1 

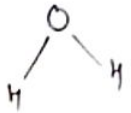
POS = 3+1 = 4

COS = X

eg 2 

POS = 7

COS = ✓

eg 3 

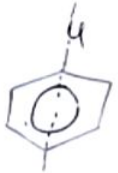
POS = 2

COS = X

eg 4 O=C=O

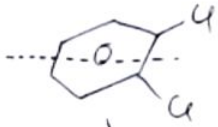
POS = ∞

COS = ✓

eg 5 

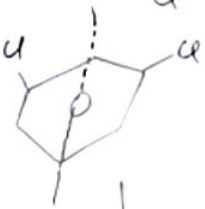
POS = 2

COS = X

eg 6 

POS = 2

COS = X

eg 7 

POS = 2

COS = X

eg 8 

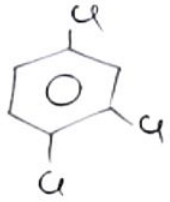
POS = 3

COS = ✓

OPTICAL ISOMERISM

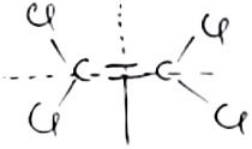
(VIII) ELEMENTS OF SYMMETRY

eg 9



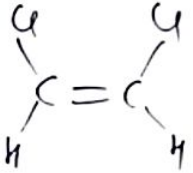
POS = 1
COS = ✗

eg 10



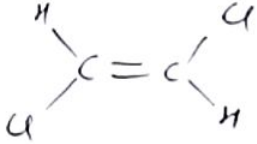
POS = 3
COS = ✓

eg 11



POS = 2
COS = ✗

eg 12



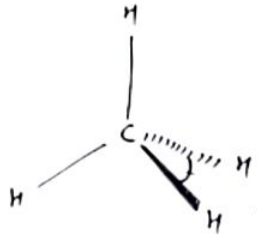
POS = 1
COS = ✓

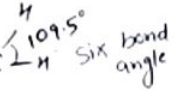
Note : All planar molecules have min. one plane of symm (always) so all planar molecules are optically inactive (achiral)

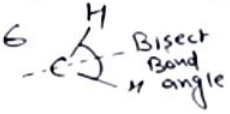
OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

eg 13

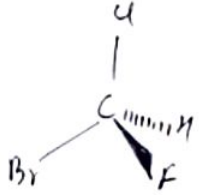


- 

- POS = 6
- 

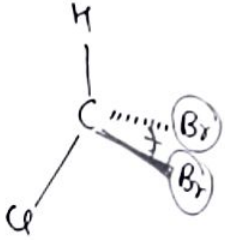
- ⇒ Cos = X

eg 15



eg 16

eg 14



POS = 1

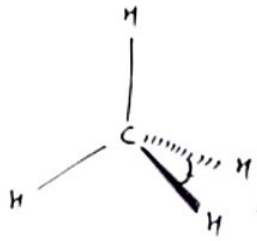
Cos = X



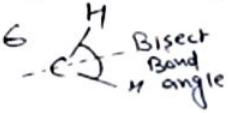
OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

eg 13



- $\angle$ 109.5° Six bond angle
- POS = 6
- COS = X



Bisect Bond angle

eg 14



POS = 1

COS = X

eg 15



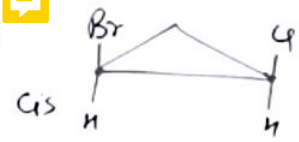
POS = X } O.A.C
COS = X } chiral

eg 16

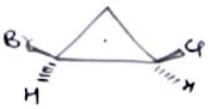
OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

Eg 16



POS = X



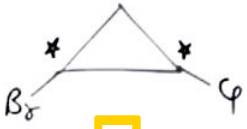
COS = X

Eg 17



POS = X

COS = X



Eg 18



meso

POS = ✓

COS = X

⇒ O.I.C

→ achiral

→ with chiral atom

Eg 19



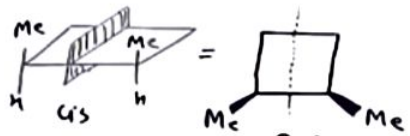
POS = X

COS = X

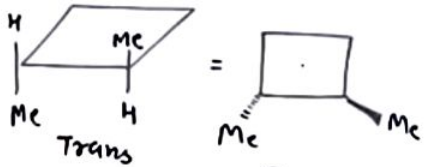


(VIII) Elements Of Symmetry

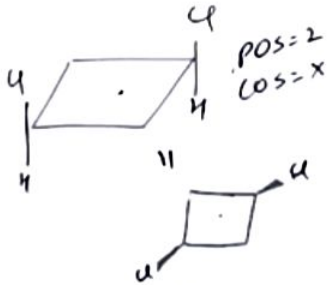
eg 20



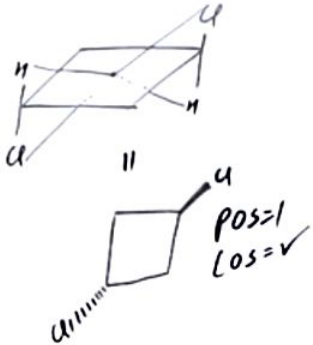
eg 21



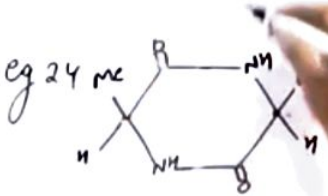
eg 22



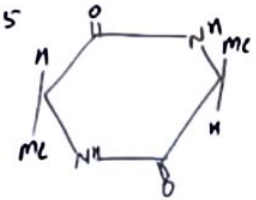
eg 23



eg 24



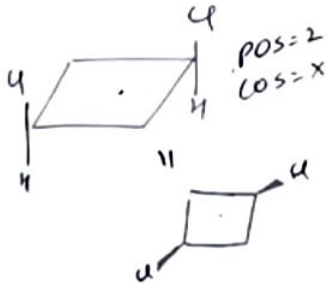
eg 25



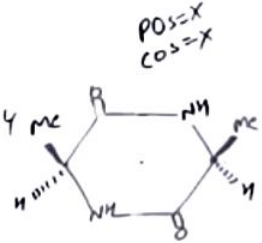
OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

eg 22



eg 24





Trans

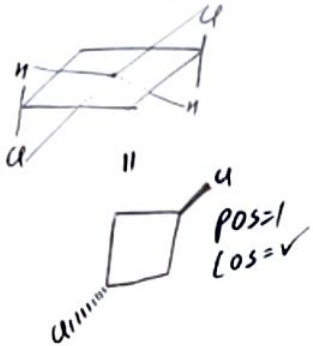
POS=X

COS=X

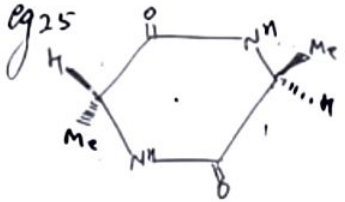
⇒ O.A.C

⇒ Chiral Comp.

eg 23



eg 25



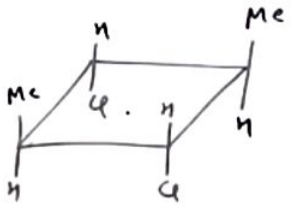
POS=X

COS=✓

OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

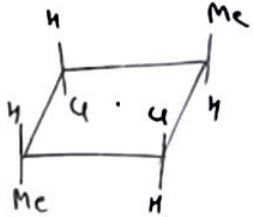
eg 26



POS = 2

COS = X

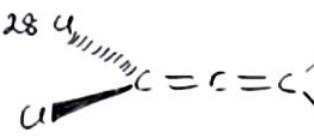
eg 27



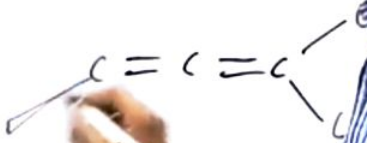
POS = X

COS = ✓

eg 28

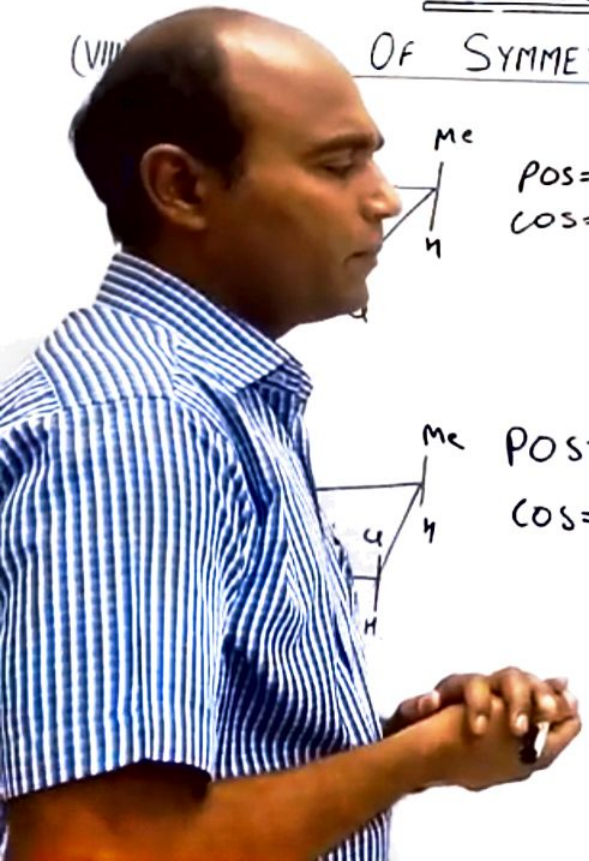


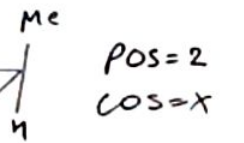
eg 29



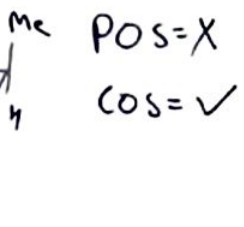
OPTICAL ISOMERISM

(VII) Of Symmetry



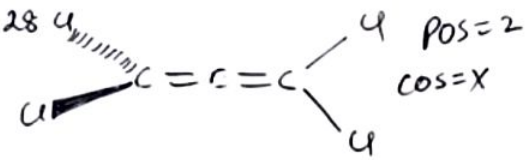


POS = 2
COS = X



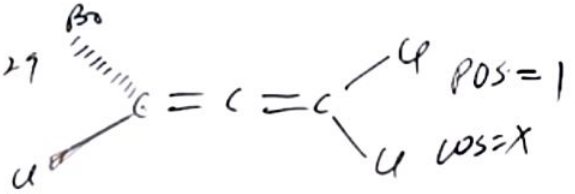
POS = X
COS = ✓

eg 28



POS = 2
COS = X

eg 29



POS = 1
COS = X

OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

eg 30



POS = X → No chiral atom
COS = X → Chiral molecule
→ O.A.C

eg 31



OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

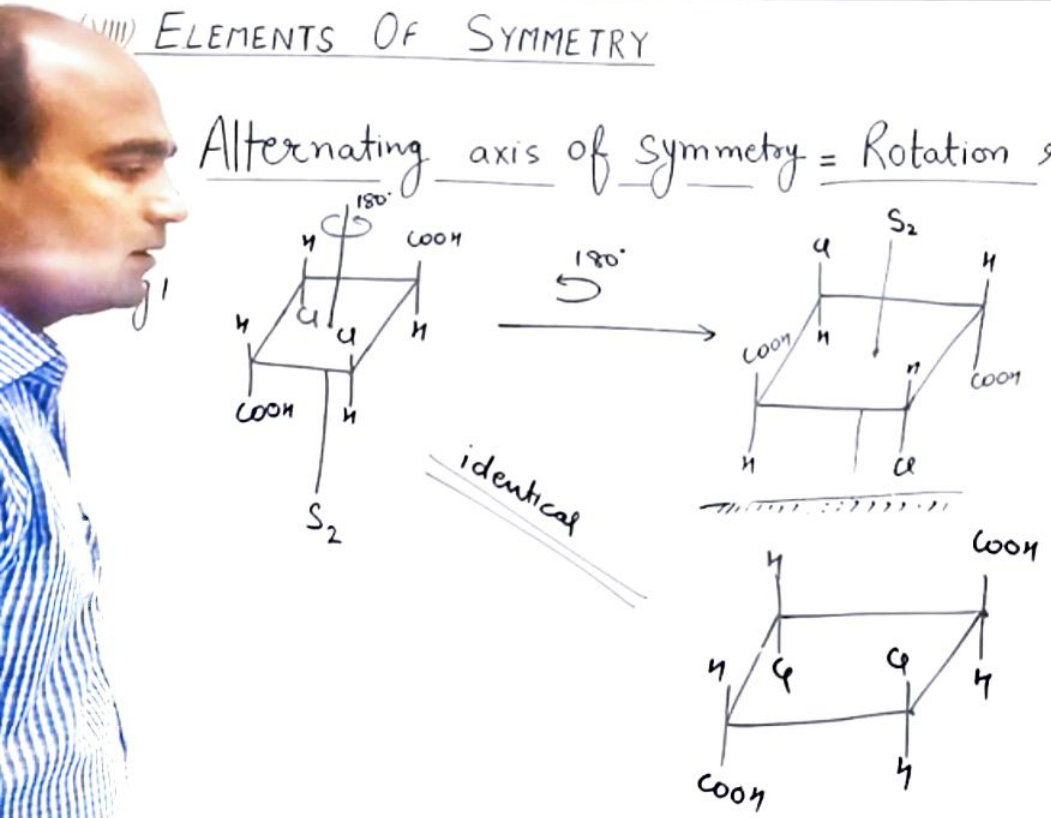
(D) Alternating axis of Symmetry = Rotation reflection axis of Symm.

A molecule passesses an n-fold of alternating axis of symmetry (S_n). When molecule rotated through an angle of $\frac{360^\circ}{n}$ about the axis and then reflected across a plane perpendicular to the axis, an identical structure is optained.

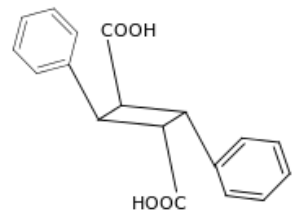
OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

Alternating axis of symmetry = Rotation reflection axis of symm.



$S_n = S_{\frac{360}{180}}$



α-truxillic acid

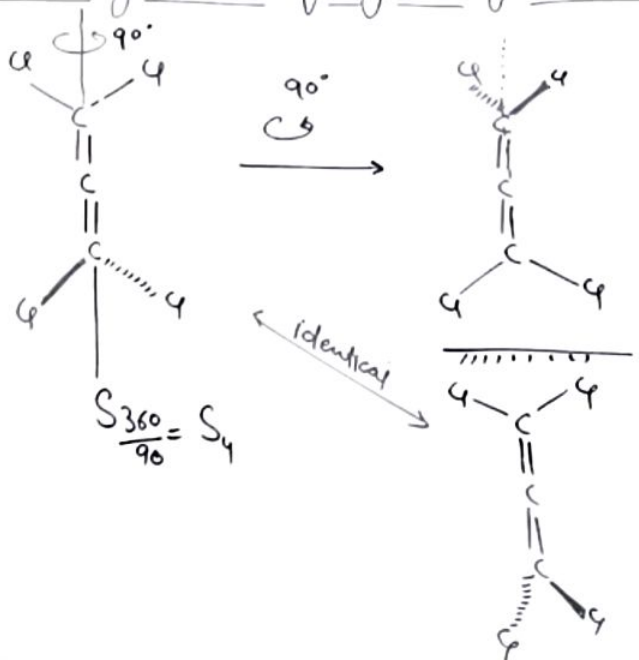
has S2 axis of symmetry

OPTICAL ISOMERISM

(VIII) ELEMENTS OF SYMMETRY

(D) Alternating axis of Symmetry = Rotation reflection

eg.





OPTICAL ISOMERISM

⇒ Asymmetric compounds:-

Compounds in which all elements of symmetry (except C_1 axis) are not exist.

⇒ Dissymmetric comp.-

Compound in which plane of symm., centre of sym. and alternating axis of symmetry are absent.

★ All asymmetric molecules are dissimmetric but its revers not always true.

OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism

[A] Enantiomers :- Stereoisomers which are mirror image of each other but they are not superimpose on each other

⇒ Enantiomers are always optically active.

⇒ Enantiomers have identical physical properties like M.P., boiling point, solubility, density etc. except the direction of rotation of light (PPL)

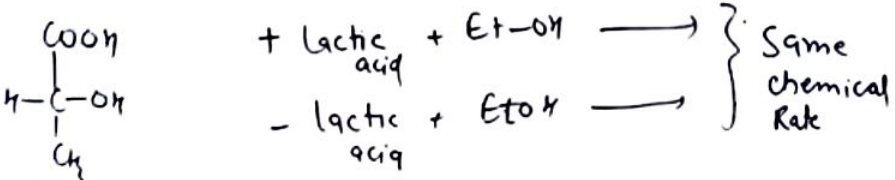
⇒ Both enantiomers can rotate the vibrational plane of PPL with same magnitude in opposite direction. (↻ or ↺)

OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism

[A] Enantiomers :-

Enantiomers have identical chemical properties with optical inactive reagent but they have different chemical property with optical active reagent.



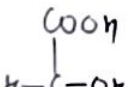
#23|OPTICAL ISOMERISM|STEREOISOMERISM|JEE MAIN, IIT ADVANCED|BYNEERAJ SAINI|NS S...

OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isom

[A] Enantiomers :-

Enantiomers have identical chemical properties w
optical inactive reagent but they have different chemica
property with optical active reagent.



Lactic acid

+ Lactic acid + d  → diff

- lactic acid + d  → diff

14:36 / 25:41

OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism

[A] Enantiomers :-

⇒ Enantiomers have identical chemical properties with optical inactive reagent but they have different chemical property with optical active reagent.

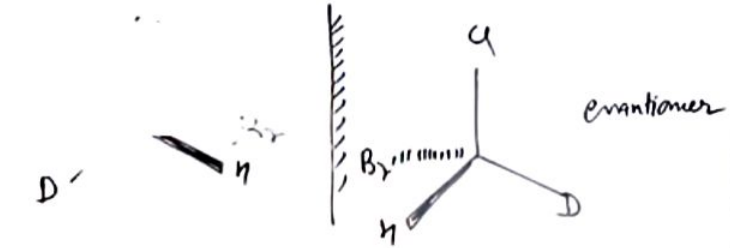
⇒ Enantiomers have different biological properties in biological System.



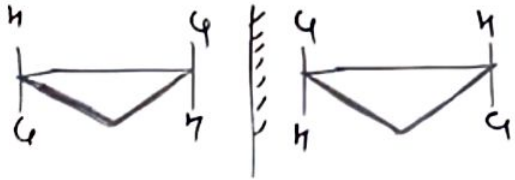
OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism

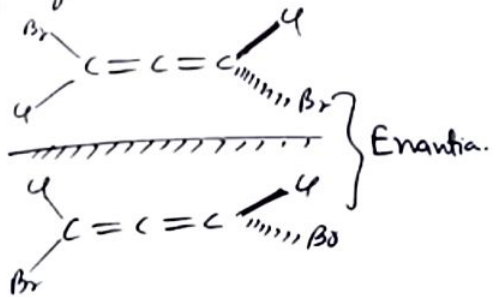
[A] Enantiomers :-



eg 2



eg 3

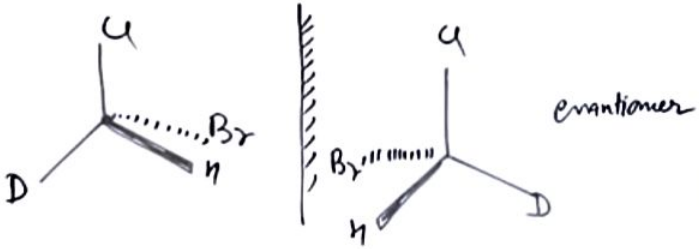


#23|OPTICAL ISOMERISM|STEREOISOMERISM|JEE MAIN, IIT ADVANCED|BYNEERAJ SAINI|NS S...

(VIII) Some important definitions related to optical isomerism

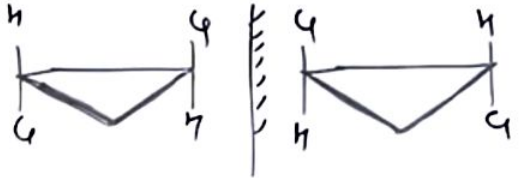
[A] Enantiomers :-

eg 1



enantiomer

eg 2



enanti·

eg 3



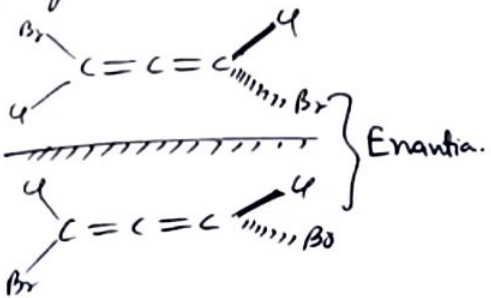
#23|OPTICAL ISOMERISM|STEREOISOMERISM|JEE MAIN, IT ADVANCED|BYNEERAJ SAINI|NS S...

OPTICAL ISOMERISM

ne important definitions related to optical isomerism

ntiomers :-

eg 3



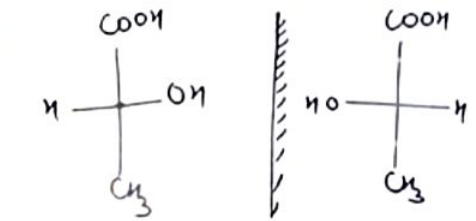
OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism

[A] Enantiomers :-

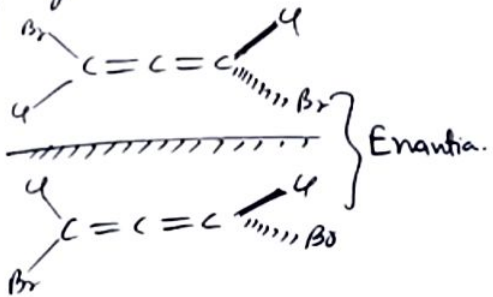
+

eg 4



lacticacid

eg 3



OPTICAL ISOMERISM

(VIII) Some Important Definitions Related to optical Isomerism

(B) **Diastereomers:-** Stereoisomers which are not mirror image of each other and non superimposable.

⇒ Diastereomers may be optical active and may not be active.

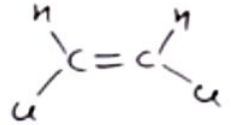
⇒ Diastereomers have different physical properties so they can be separated by normal physical separation method. (like fractional distillation, recrystallisation)

OPTICAL ISOMERISM

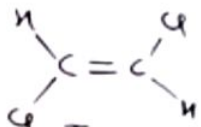
(VIII) Some Important Definitions Related to optical Isomerism.

(B) Diastereomers:-

eg 1



Cis



Trans

geometrical diastereomers

eg 2



Cis



Trans.

dias

eg 3





diastereomers.

eg 4



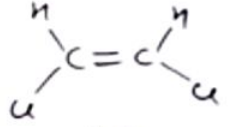


OPTICAL ISOMERISM

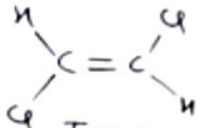
(VIII) Some Important Definitions Related to optical Isomerism

(B) Diastereomers:-

eg 1

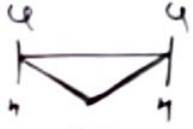


Cis



Trans

geometrical diastereomers



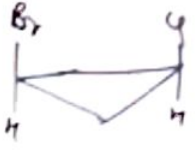
Cis



Trans.

dias

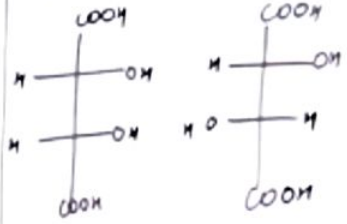
eg 3





diastereomers.

eg 4



Diastereomers

OPTICAL ISOMERISM

(VIII) Some Important Definitions Related to optical Isomerism

(c) Meso Compounds :-

Chiral atom containing optical inactive compounds are meso compounds

- Meso compounds are optically inactive due to internal compensation

(VIII) Some Important Definitions Related to optical Isomerism

(c) Meso Compounds:-

eg 1



POS = ✓

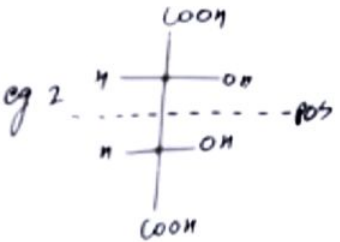
- achiral molecule

⇒ Optical inactive Comp.

⇒ Chiral atom = ✓

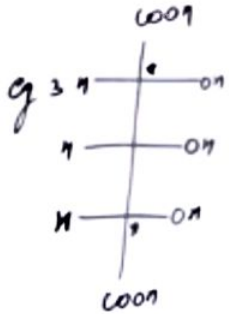
⇒ Meso

eg 2



Meso Tartaric acid

eg 3



meso = ✓

Optical inact...

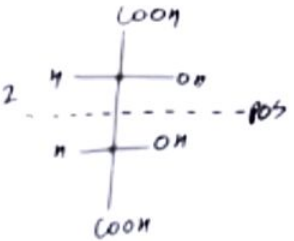
#24|OPTICAL ISOMERISM|STEREOISOMERISM|JEE MAIN, IIT ADVANCED|BYNEERAJ SAINI|NS S...

OPTICAL ISOMERISM

(VIII) Some Important Definitions Related to optical Isomerism

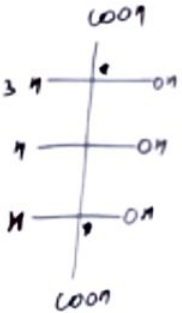
(c) Meso Compounds:-

eg 2



Meso Tartaric acid

eg 3



meso = ✓

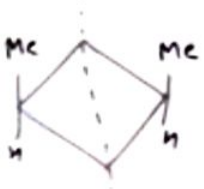
Optical inactive

OPTICAL ISOMERISM

(VIII) Some Important Definitions Related to optical Isomerism

(c) Meso Compounds :-

eg 4



Pos = ✓

Optical inactive

⇒ No Chiral atom

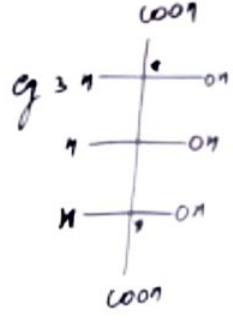
⇒ Not meso

eg 5



Meso

eg 3



meso = ✓

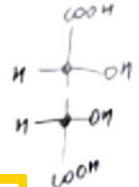
Optical inactive

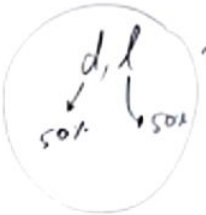
OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism

(1) Racemic mixture = Racemic modification = (±) = dl

- A mixture which contain equal amount of enantiomers.
- A pure racemic mixture is optical inactive due to presence of external compensation





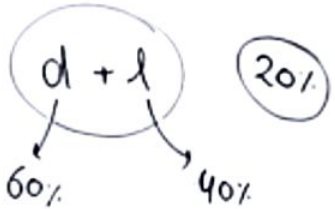
OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism:-

(1) Racemic mixture = Racemic modification = (±) = dl

- A mixture which contain equal amount of enantiomers.
- A pure racemic mixture is optical inactive due to presence of external compensation

⇒ Enantiomeric excess (ee):-



OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism:-

(D) Racemic mixture = Racemic modification = (±) = dl

- A mixture which contain equal amount of enantiomers.
- A pure racemic mixture is optical inactive due to presence of external compensation

⇒ **Enantiomeric excess (ee)**: When enantiomers are present in unequal amount in a mixture, then excess % of predominant enantiomer is called enantiomeric excess. Its value is equals to value of optical purity for mixture.

OPTICAL ISOMERISM

I) Some important definitions related to optical isomerism:-

$$e.e = \frac{|d-l|}{d+l} \times 100 = \text{optical purity} = \frac{[\alpha]_{mix}}{[\alpha]_{pure}} \times 100$$



OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism:-

Ques Identify values of $[\alpha]_{mix}$ for given compositions

if $[\alpha]_{pure} = +80^\circ$

$$[\alpha]_{mix} = \frac{ee}{100} \times [\alpha]_{pure} = \frac{20}{100} \times 80^\circ$$

	%d	%l	e.e.	% Racemic mix.	$[\alpha]_{mix}$
①	50%	50%	0	100%	0°
②	100%	0%	100%	0%	$+80^\circ$
③	60%	40%	20%	80%	$+16^\circ$
④	80%	20%	60%	40%	$+48^\circ$
⑤	25%	75%	50%	50%	-40°
⑥	33.33%	66.67%	33.34%	66.66%	$-\frac{80}{3} = 26.6^\circ$

OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism:-

Ques Identify values of $[\alpha]_{mix}$ for given compositions

if $[\alpha]_{pure} = +80^\circ$

$$[\alpha]_{mix} = \frac{ee}{100} \times [\alpha]_{pure} = \frac{20}{100} \times 80^\circ$$

%d	%l	e.e.	% Racemic mix	$[\alpha]_{mix}$
(50+15)% 65%	(50-15)% 35%	30%	70%	$+24^\circ$
(50-20)%	(50+20)%	40%	60%	-32°
80%	20%	60%	40%	$+48^\circ$
25%	75%	50%	50%	-40°
33.33%	66.67%	33.34%	66.66%	$-\frac{80}{3} = 26.6^\circ$

$$ee = \frac{[\alpha]_{mix}}{[\alpha]_{pure}} \times 100$$

$$= \frac{24}{80} \times 100 = 30\%$$

$\Rightarrow d - l = 30$

$d + l = 100$

OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism:-

(E) Resolution:-

- We know Enantiomers can not be separated by normal physical separation method from a racemic mix. because enantiomers have identical physical properties.

→ Seperation method of enanantiomer from a mixture of enantiomers is called Resolution.

⇒ Enantiomers are also known as resolvable compound.

OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism :-

(E) Resolution

- Mechanical resolution
- Biological resolution
- Chromatography resolution
- Chemical method = Diastereomeric method.

OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism:-

Diastereomeric resolution:-

In this method racemic mixture is treated with pure optical active comp. (reactant) form mixture of diastereomeric products these diastereomeric products are isolable (by physical separation method). this isolated diastereomer may be regenerate and isolate parent enantiomer.

OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism:-

Diastereomeric resolution:-

$\underset{\text{Racemic mix.}}{(A_{(R)} + A'_{(S)})_{mix}}$ + $\underset{\text{pure optical active comp.}}{B_{(R)}}$ ⟶ $[A_{(R)}-B_{(R)} + A'_{(S)}-B_{(R)}]_{mix.}$ mixture of diastereomers (optical active mix.)

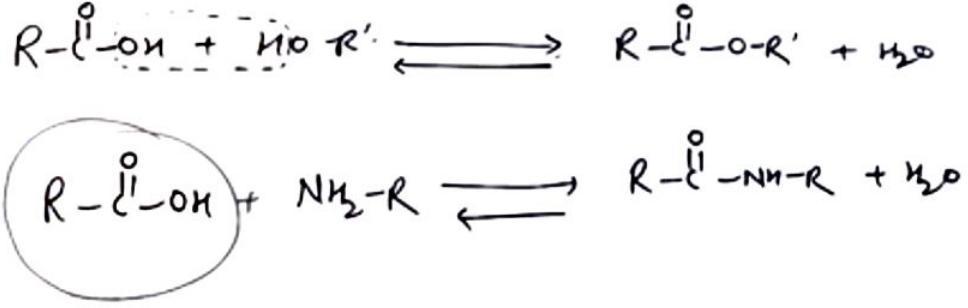
Fractional distillation

- $A_{(R)}-B_{(R)}$ → Chemical Rx^n → $A_{(R)} + B_{(R)}$
- $A'_{(S)}-B_{(R)}$ → Chemical Rx^n → $A'_{(S)} + B_{(R)}$

OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism:-

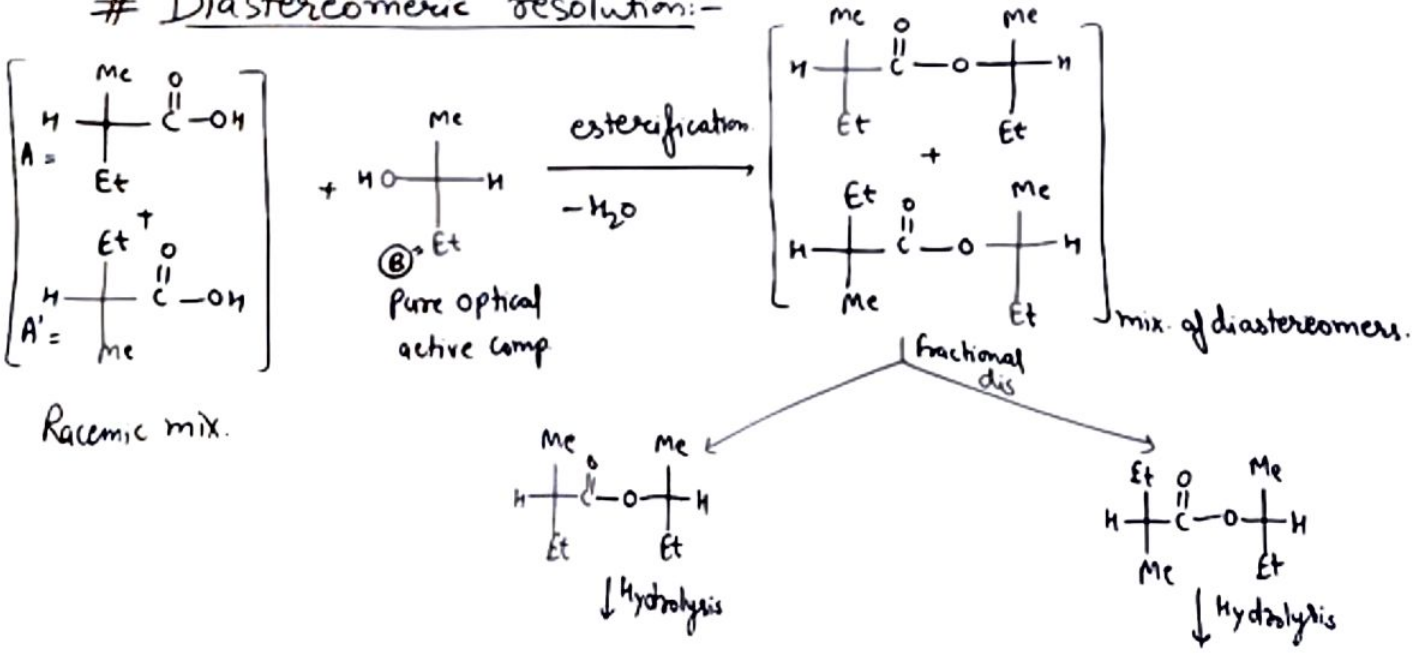
Diastereomeric resolution:-



OPTICAL ISOMERISM

(VIII) Some important definitions related to optical isomerism:-

Diastereomeric resolution:-



#27|OPTICAL ISOMERISM|STEREOISOMERISM|JEE MAIN, IIT ADVANCED|BYNEERAJ SAINI|NS S...

OPTICAL ISOMERISM

(IX) Three Dimensional Representation Of Molecules :-

(3D - Projection formula)

- (A) → Wedge - dash projection formula
- (B) → Fischer " "
- (C) → Sawhorse " "
- (D) → Newmann " "

} not applicable for only one chiral carbon.

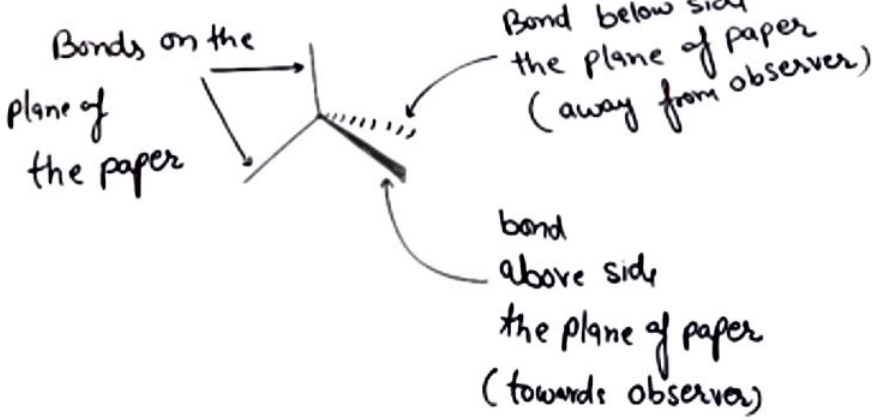


OPTICAL ISOMERISM

(IX) Three Dimensional Representation Of Molecules :-

[A] Wedge - dash projection formula :-

» For one chiral carbon.



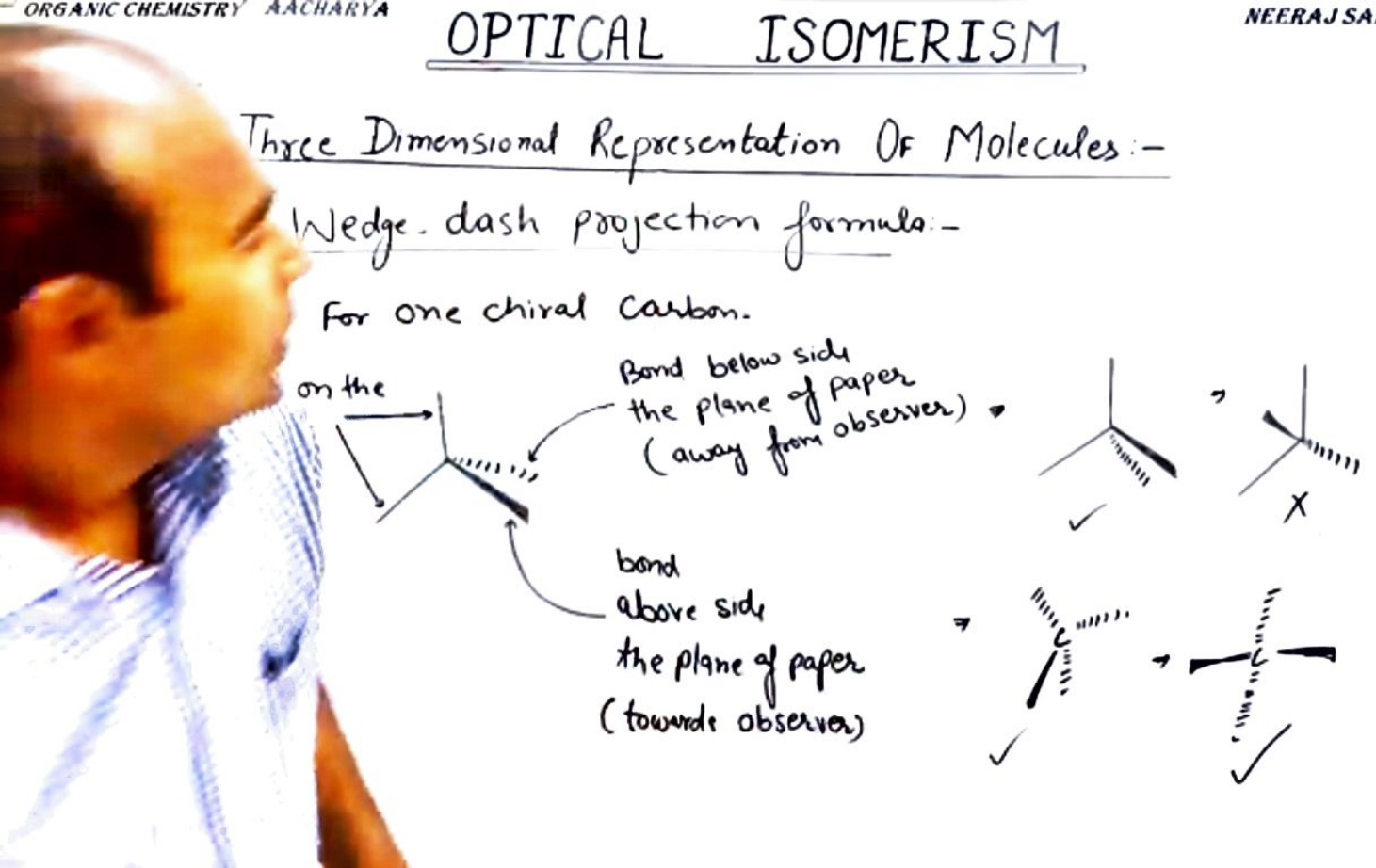


OPTICAL ISOMERISM

Three Dimensional Representation Of Molecules :-

Wedge - dash projection formula :-

For one chiral carbon.

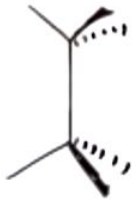


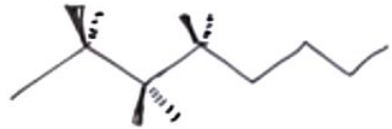
OPTICAL ISOMERISM

(IV) Three Dimensional Representation Of Molecules :-

[A] Wedge - dash projection formula :-

- For more chiral carbons :

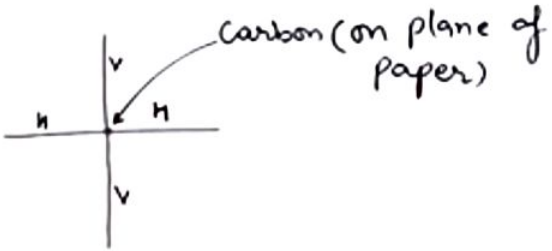


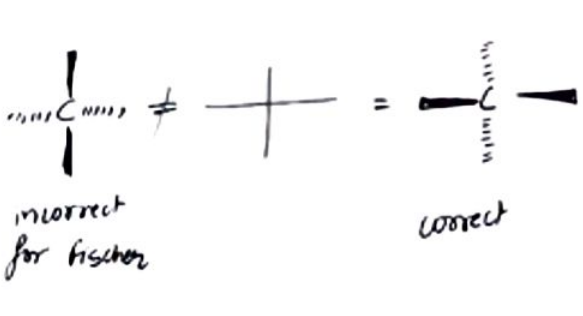


OPTICAL ISOMERISM

(IX) Three Dimensional Representation Of Molecules :-

[B] Fischer projection formula -





- Horizontal lines = Bonds above side of plane of paper (Towards observer)
- Vertical lines = Bond below side of plane of paper (away from observer)



OPTICAL ISOMERISM

(IX) Three Dimensional Representation Of Molecules :-

[B] Fischer projection formula :-

1. We obtain identical structure by 180° rotation of fischer formula in the plane of paper.
2. We obtain enantiomeric structure by 90° rotation of fischer formula in the plane of paper.
3. We obtain identical configuration on chiral carbon by even no. of simultaneous exchanges of group.

#28|OPTICAL ISOMERISMISTEREOISOMERISM|JEE MAIN, IT ADVANCED|BYNEERAJ SAINI|NS S...

(IX) Three Dimensional Representation Of Molecules :-

[B] Fischer projection formula :-

④ We get inversion in configuration of chiral carbon by odd no. of simultaneous exchanges of groups on same chiral carbon.

2 We obtain enantiomeric structure by 90° rotation of fischer

th

hiral carbon by even no. of

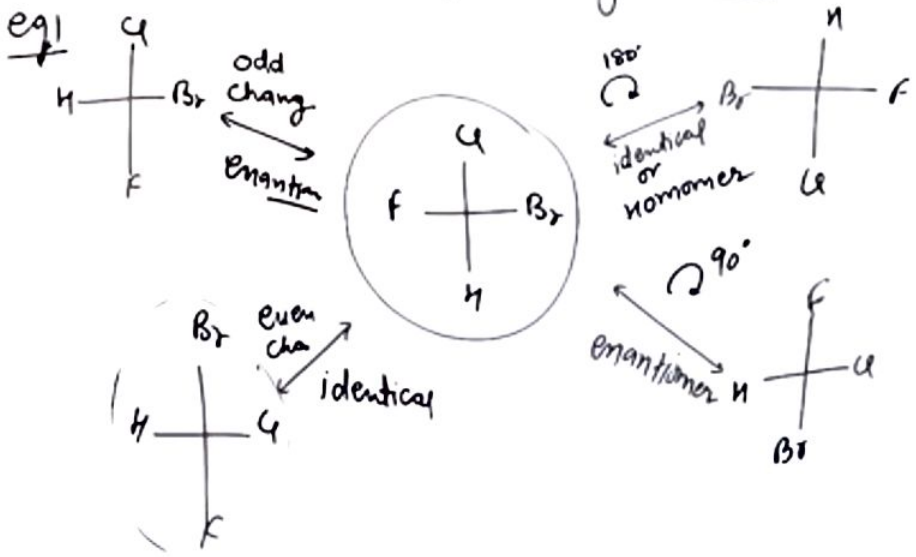
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OPTICAL ISOMERISM

(IX) Three Dimensional Representation Of Molecules :-

[B] Fischer projection formula :-

eg1



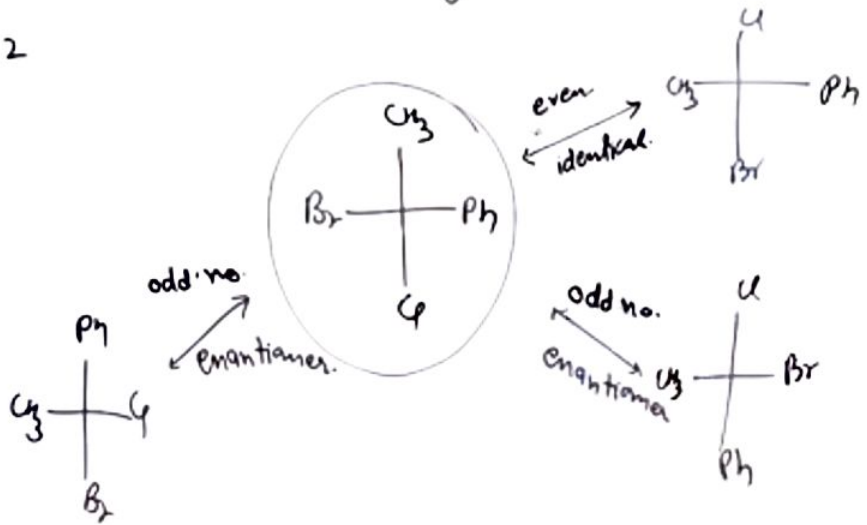


OPTICAL ISOMERISM

(IX) Three Dimensional Representation Of Molecules

[B] Fischer projection formula :-

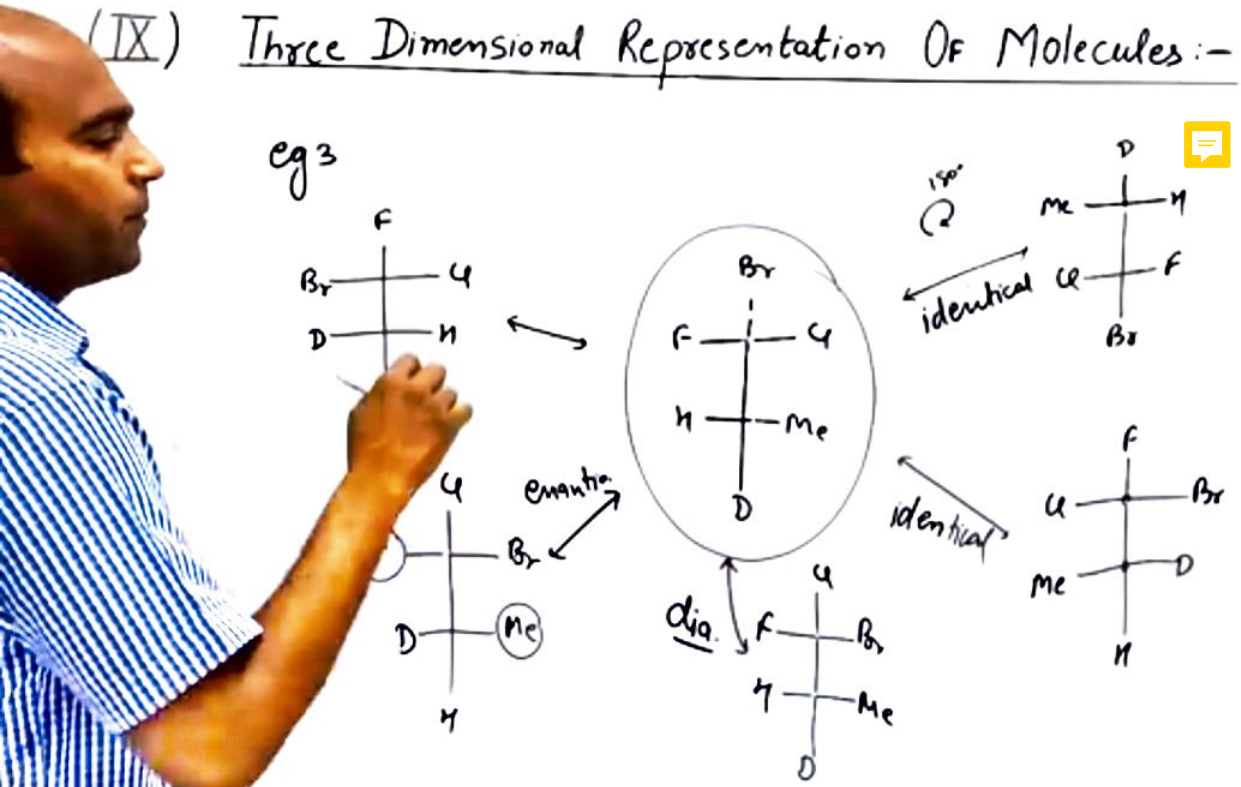
Eg 2





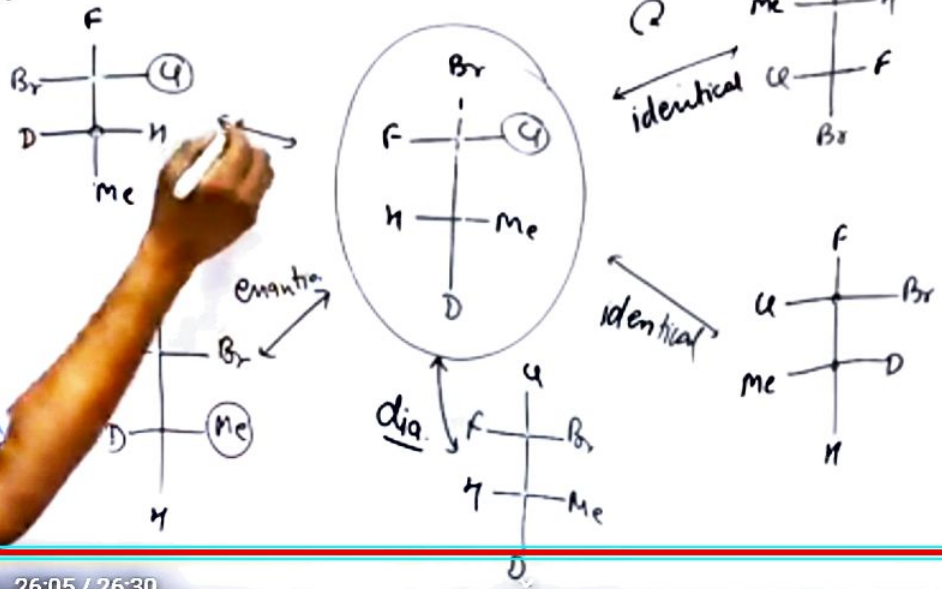
OPTICAL ISOMERISM

(IX) Three Dimensional Representation Of Molecules :-



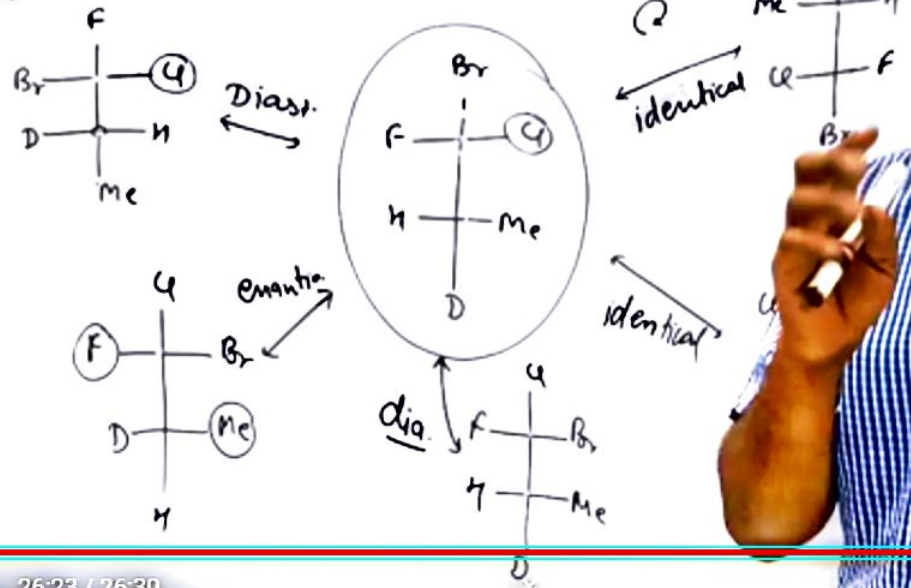
(IV) Three Dimensional Representation Of Molecules :-

eg 3



(IX) Three Dimensional Representation Of Molecules :-

eg 3

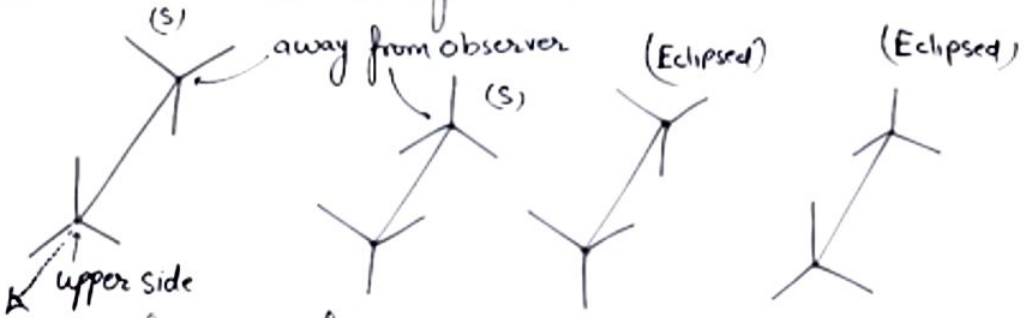


NEERAJ SAINI (N.S. SIR)

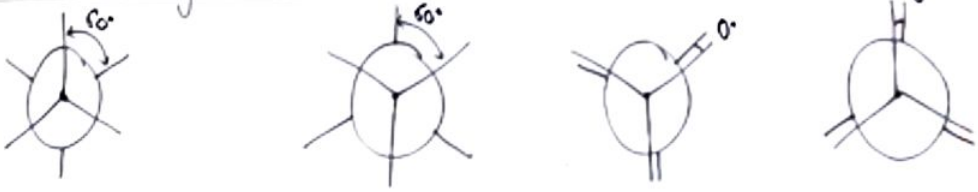
OPTICAL ISOMERISM

(IX) Three Dimensional Representation Of Molecules :-

(C) Sawhorse projection formula :-



(D) Newmann's projection formula :



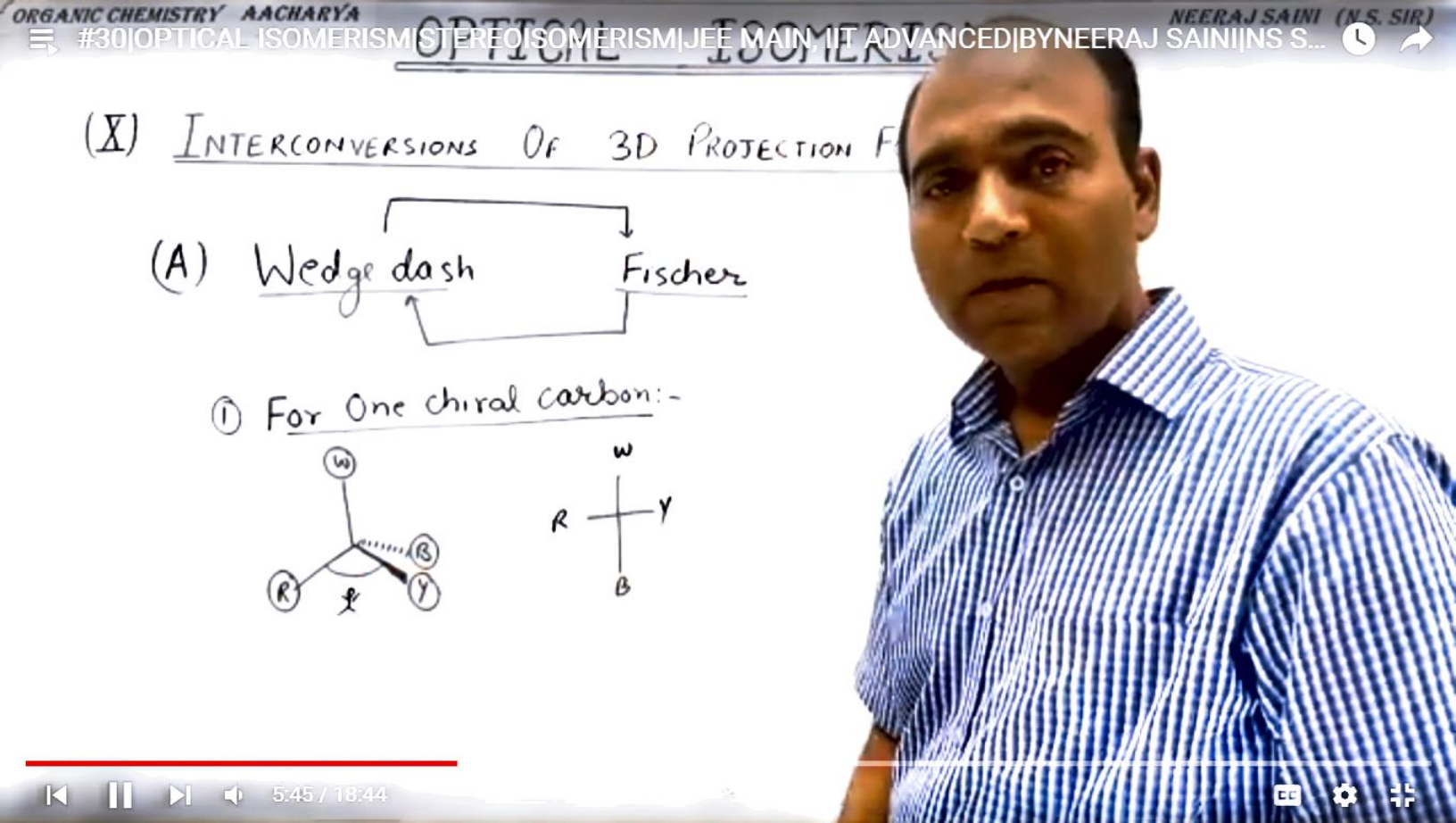
#30|OPTICAL ISOMERISM|STEREOISOMERISM|JEE MAIN, IIT ADVANCED|BYNEERAJ SAINI|NS S...

OPTICAL ISOMERISM

(X) INTERCONVERSIONS OF 3D PROJECTION F

(A) Wedge dash ⇄ Fischer

① For One chiral carbon:-

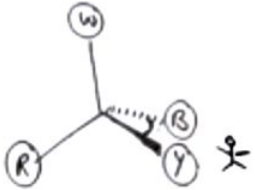


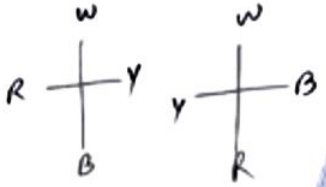
OPTICAL ISOMERISM

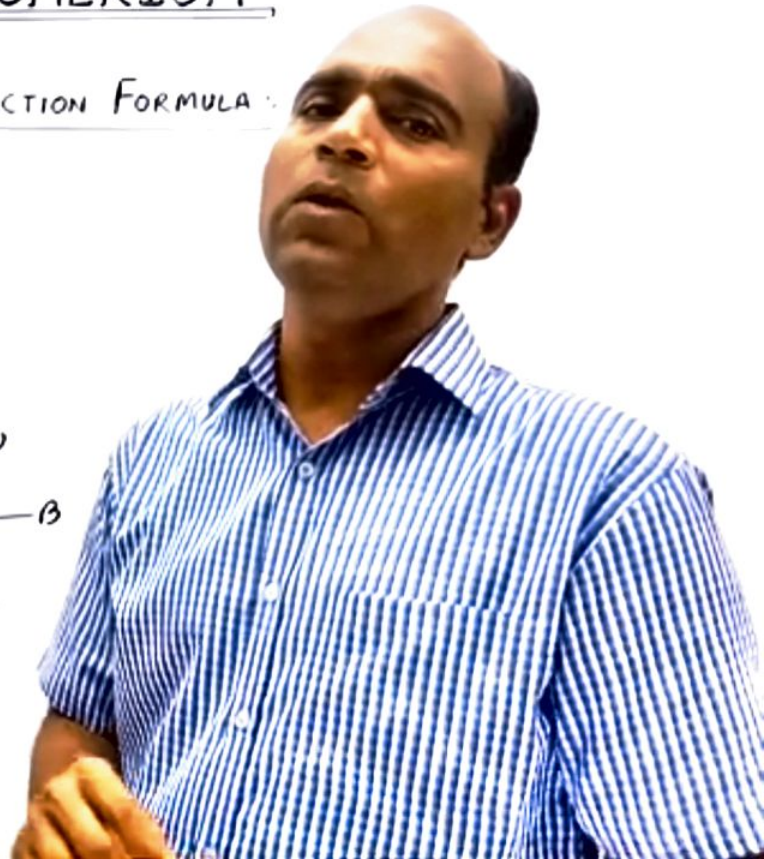
(X) Interconversions Of 3D Projection Formula :

(A) Wedge dash ⇄ Fischer

① For One chiral carbon :-





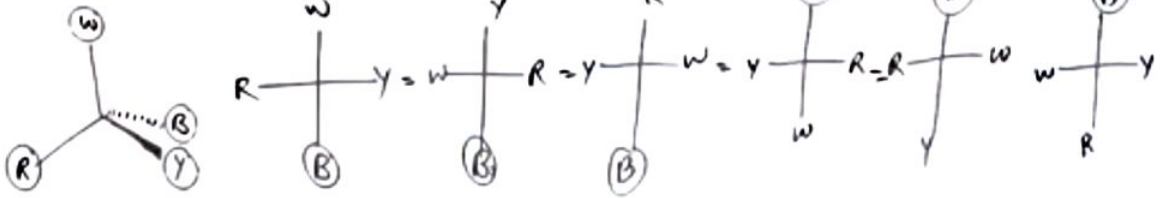


OPTICAL ISOMERISM

(X) Interconversions Of 3D Projection Formula:

(A) Wedge dash ⇄ Fischer

(1) For One chiral carbon:-



Short trick

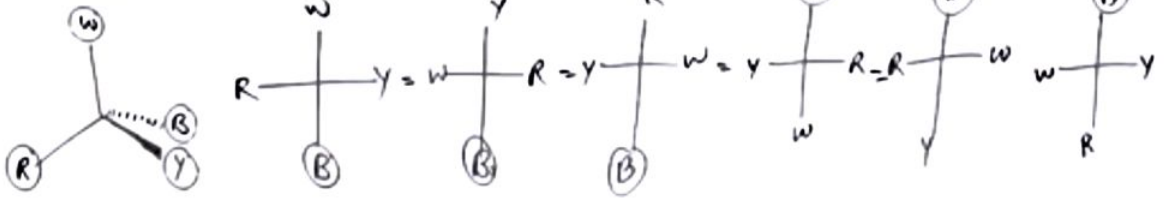
- ⇒ Dotted group on vertical line
- ⇒ Remaining group in same sequence on remaining position

OPTICAL ISOMERISM

(X) Interconversions Of 3D Projection Formula :

(A) Wedge dash ⇄ Fischer

① For One chiral carbon :-



Short trick

→ Dotted group on vertical line

→ Remaining group in same sequence on remaining position

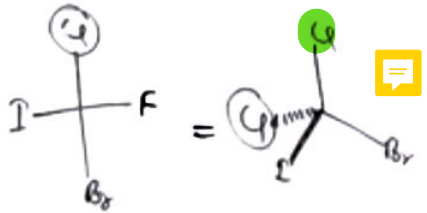
#30|OPTICAL ISOMERISM|STEREOISOMERISM|JEE MAIN, IT ADVANCED|BYNEERAJ SAINI|NS S...

OPTICAL ISOMERISM

(X) Interconversions Of 3D Projection Formula :

(A) Wedge dash ⇄ Fischer

① For One chiral carbon :-



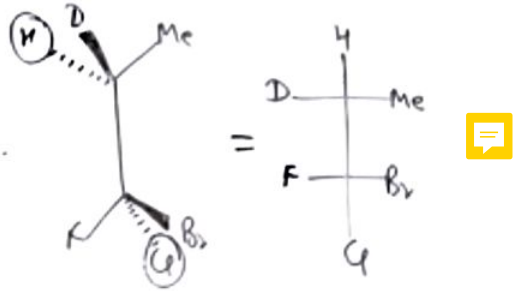
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OPTICAL ISOMERISM

(X) Interconversions Of 3D Projection Formula :

(A) Wedge dash ⇄ Fischer

(2) For Two Chiral Carbons



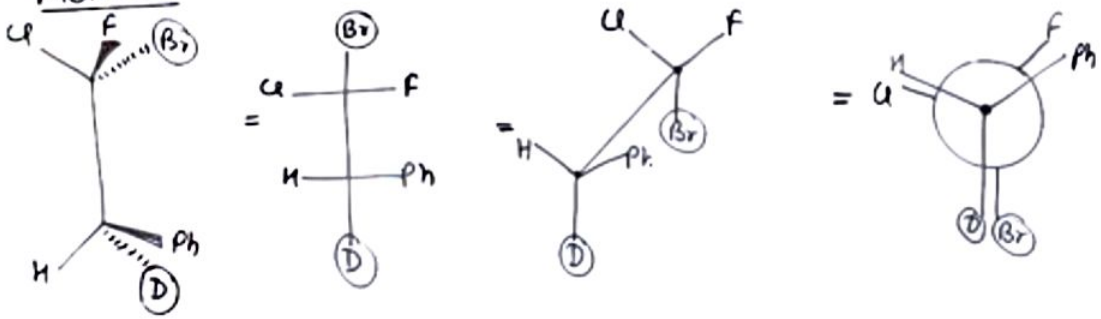


OPTICAL ISOMERISM

(X) Interconversions Of 3D Projection Formula :

(B) Wedge dash, Fischer, Newmann's and Sawhorse

Method I

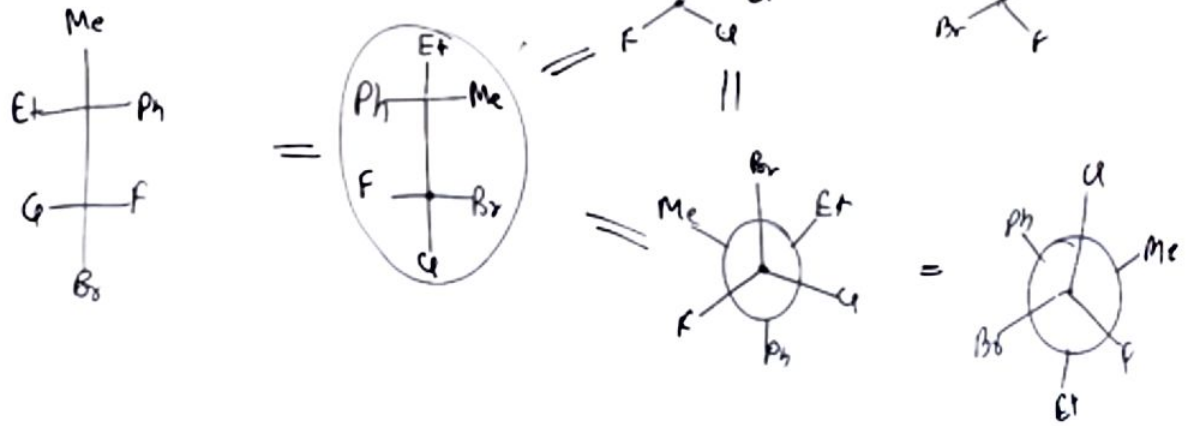


OPTICAL ISOMERISM

(X) Interconversions Of 3D Projection Formula :

(B) ~~Wedge~~ dash, Fischer, Newmann's and Sawhorse

Method-2



OPTICAL ISOMERISM

(X) Interconversions Of 3D Projection Formula :-

(B) ~~Wedge~~ dash, Fischer, Newmann's and Sawhorse

Method.2 For formation of identical compounds.

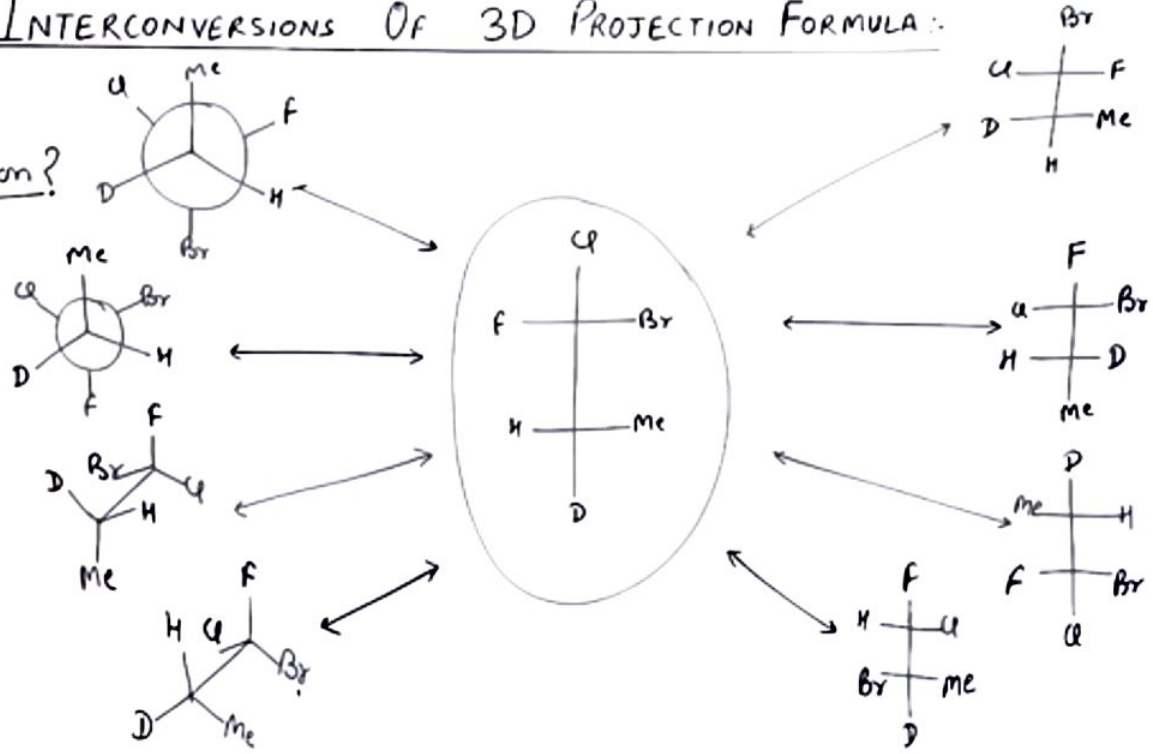


Interconversion	Sequence on Front Side carbon	Sequence on back Side carbon
① Fischer ↔ Fischer	Same sequence	Same sequence
② Fischer ↔ Sawhorse	" "	Opposite "
③ Fischer ↔ Newmann	" "	Opposite "
④ Sawhorse ↔ Sawhorse	" "	Same "
⑤ Newmann ↔ Newmann	" "	Same "
⑥ Sawhorse ↔ Newmann	" "	same "

OPTICAL ISOMERISM

(X) INTERCONVERSIONS OF 3D PROJECTION FORMULA :

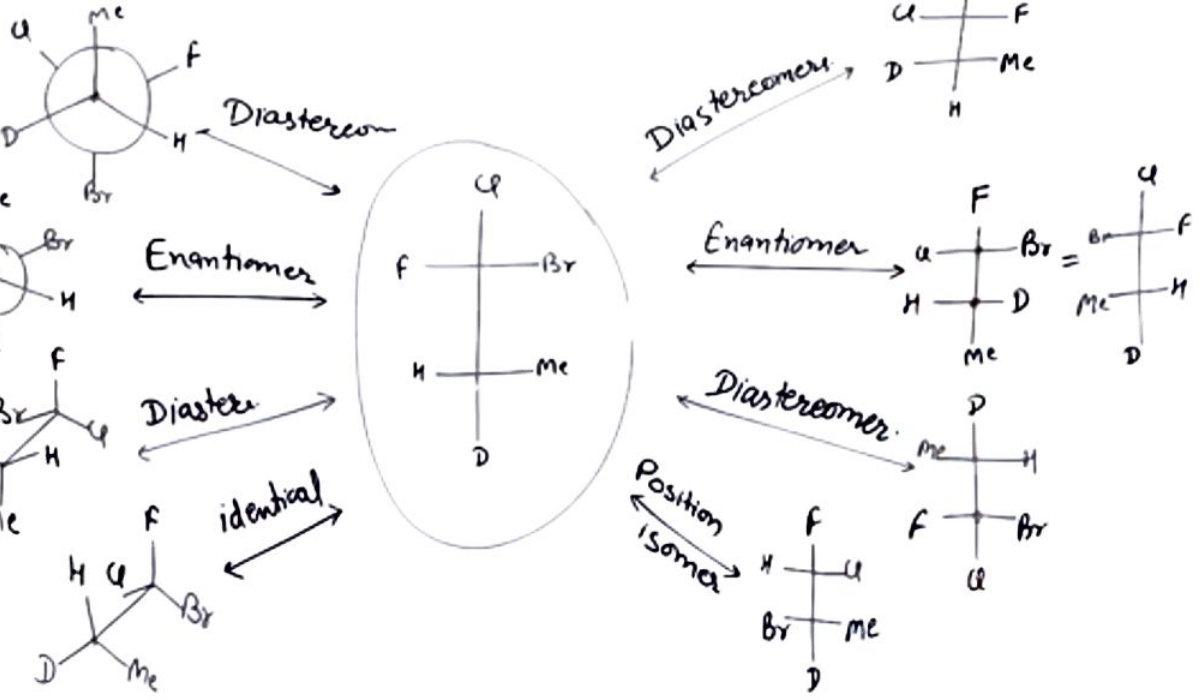
Ques : Relation ?



OPTICAL ISOMERISM

(X) INTERCONVERSIONS OF 3D PROJECTION FORMULA :

Ques : Relation?



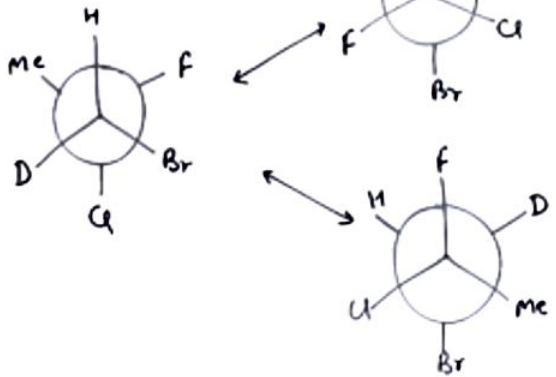
#32|OPTICAL ISOMERISM|STEREOISOMERISM|JEE MAIN|IIT ADVANCED|BYNEERAJ SAINI|NS S...

OPTICAL ISOMERISM

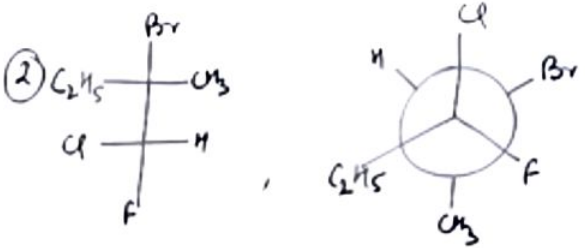
(X) Interconversions Of 3D Projection Formula :-

Ques :- Identify Relation

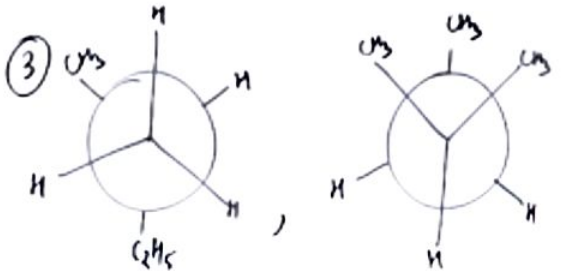
①



②



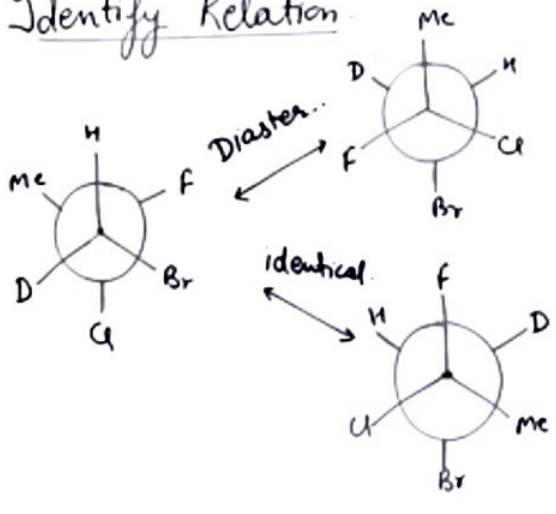
③

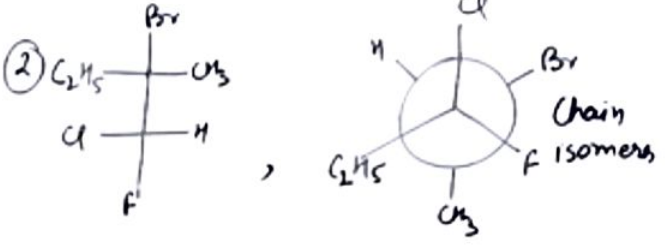


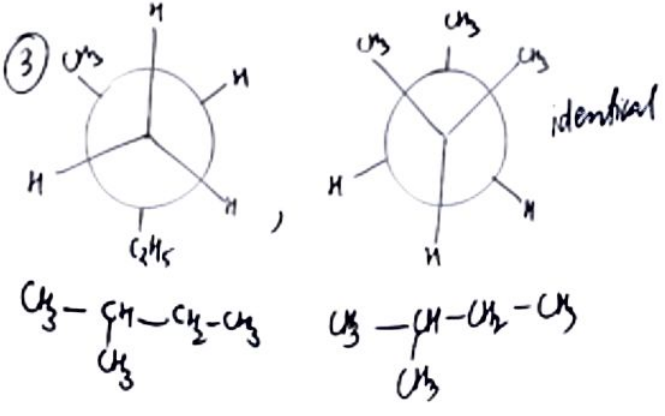
OPTICAL ISOMERISM

(X) INTERCONVERSIONS OF 3D PROJECTION FORMULA :-

Ques :- Identify Relation

① 

② 

③ 

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons :-

(A) Relative configuration
- i) → D-L Nomenclature
- ii) → Threo - Erythro

(B) Absolute configuration
- → R-S Nomenclature

A.i = D-L Nomenclature :-

This nomenclature is applicable for carbohydrates, amino acids and there derivatives.

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons :-

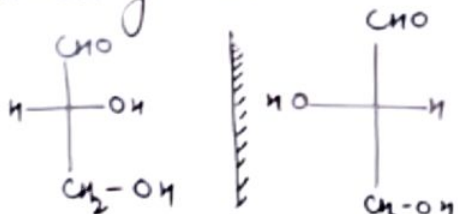
- Carbohydrates :- Poly hydroxy carbonyl compounds
- Amino acid :- α. Amino carboxylic acid
- For identification of D-L Nomenclature draw Fischer formula of given compound in which first carbon (most oxidised carbon) present on top of vertical line and last carbon on bottom of vertical line. then compare with reference compound.

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

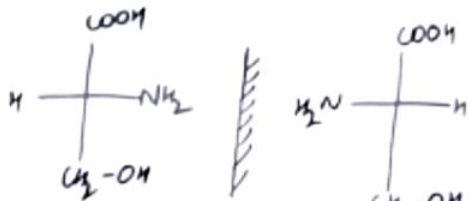
Reference compounds

For carbohydrate



D-(+)-Glyceraldehyde L(-) Glyceraldehyde

for Amino acids



D(+) Serine L(-) Serine

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

For Carbohydrate:-

When –OH group of highest numbered chiral carbon is present on RHS then configuration will = D series.

LHS -- L

For Amino acids:-

When $-NH_2$ group of α-position of –COOH group present on RHS ⇒ D

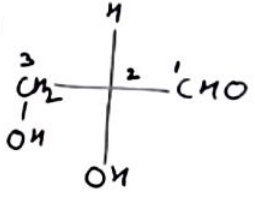
LHS ⇒ L

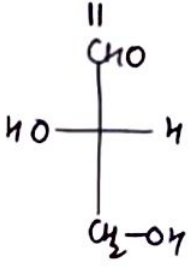


OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

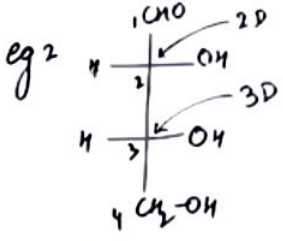
eg 1





L Series

eg 2

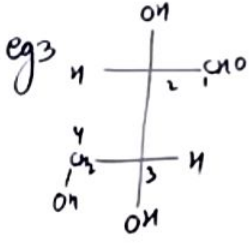


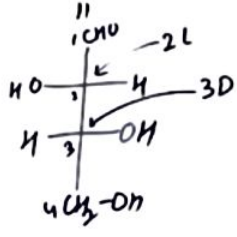
D. Series

D - Erythrose

(2D, 3D) 2,3,4-Trihydroxy-butanal

eg 3



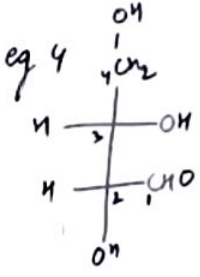


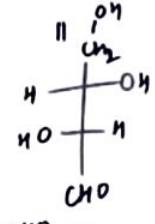
D. Series

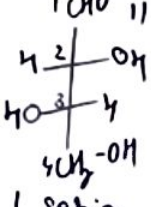
D - Threose

(2L, 3D) .....

eg 4





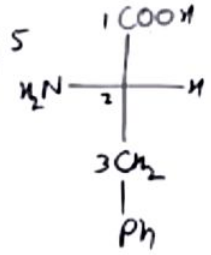


L Series

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

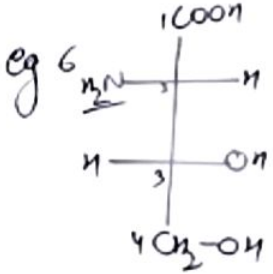
eg 5



L series (amino acid)

L. Phenyl alanine

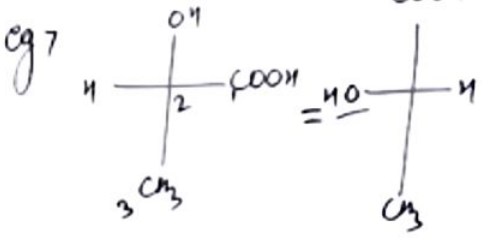
eg 6



L. Amino acid

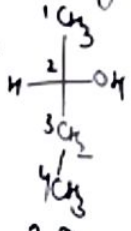
L. Threonine

eg 7



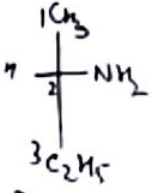
L - Lactic acid

eg 8



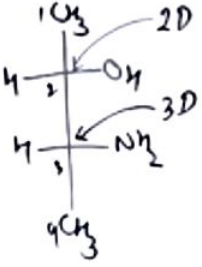
2-D

eg 9



D-amine

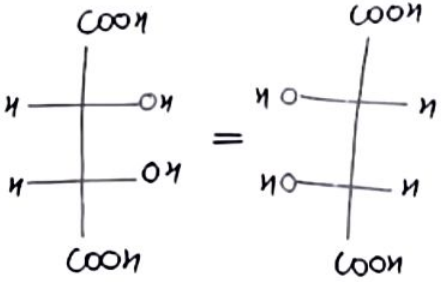
eg 10



OPTICAL ISOMERISM

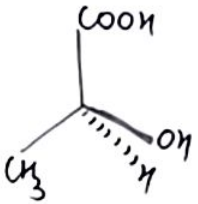
(XI) Configurational Nomenclature of Chiral Carbons:-

eg 11

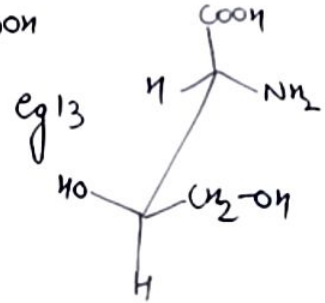


D-L Nomenclature is not applicable for similar ends containing compounds.

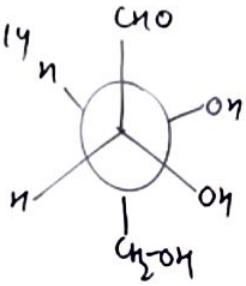
eg 12



eg 13

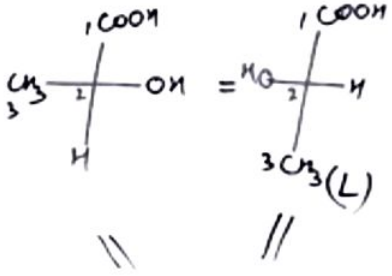


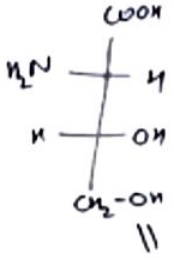
eg 14

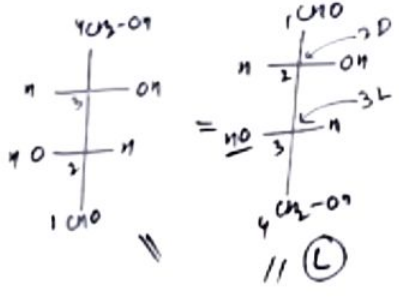


OPTICAL ISOMERISM

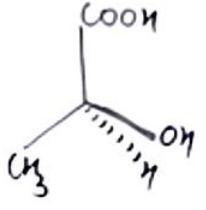
(XI) Configurational Nomenclature of Chiral Carbons:-





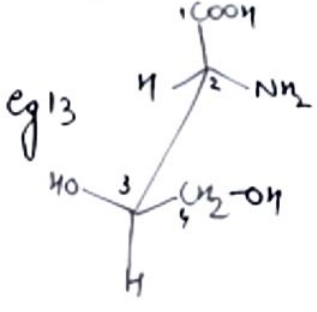


eg 12



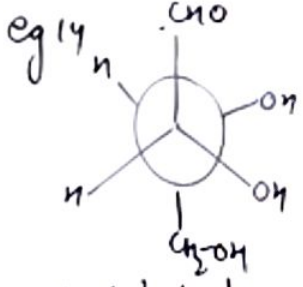
L. Lactic acid

eg 13



$L_{(S)}$ amino acid

eg 14



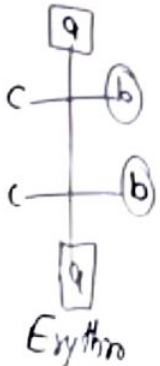
L Carbohydrate

OPTICAL ISOMERISM

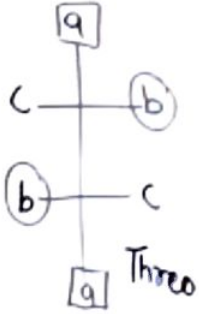
(XI) Configurational Nomenclature of Chiral Carbons:-

A2 Threo - Erythro Nomenclature:-

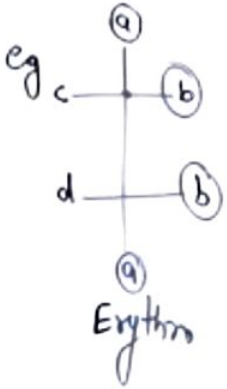
This nomenclature is applicable for 2 chiral carbon containing compounds in which atleast 2-2 groups should same of both chiral carbon.



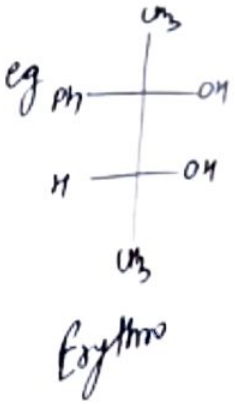
Erythro



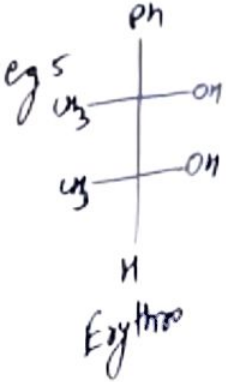
Threo



Erythro



Erythro



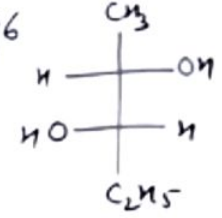
Erythro

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

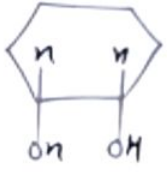
A.2 Threo - Erythro Nomenclature:-

eg6



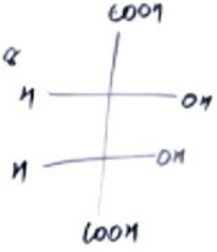
Threo

eg7



Erythro

eg8



Erythro

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

(B) Absolute configuration:-

R-S. Nomenclature:-

R = Rectus = Right

S = Sinister = left

For identification of R-S Nomenclature of chiral carbon, give priority order of four different group of chiral carbon according to 'CIP Sequence Rule' and visualise the molecule in such a way that the lowest priority group (4th) group is directed away from the

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

observer and remaining three (1, 2, 3) groups are facing to observer

if rotation of 1 → 2 → 3 group ↻ (clock wise rotation) = R

" " " " " " ↺ (anti clock wise ") = S

For identification of K-S Nomenclature of chiral carbon, give priority order of four different group of chiral carbon according to 'CIP Sequence Rule' and visualise the molecule in such a way that the lowest priority group (4th) group is directed away from the

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

Short Trick for R-S Nomenclature

- Draw Fischer projection formula of given compound
- Give priority order of four groups of Chiral carbon according to CIP sequence rule
  - ⇒ When 4th group is present on vertical line and rotation of $1 \to 2 \to 3$ ↻ (Clock wise rotation) ⇒ R
    and $1 \to 2 \to 3$ ↺ (anticlock wise rotation ⇒ S
- When 4th priority group on horizontal line
  and $1 \to 2 \to 3$ ↻ = S or $1 \to 2 \to 3$ ↺ = R

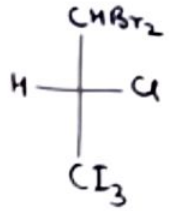
OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

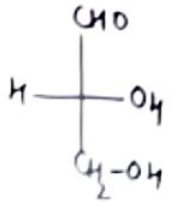
eg 1



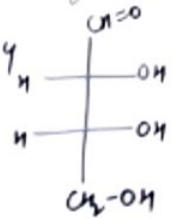
eg 2



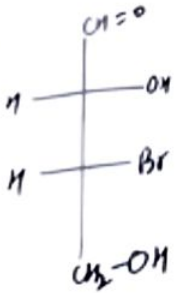
eg 3



eg 4



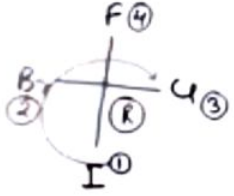
eg 5



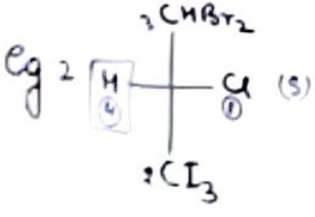
OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

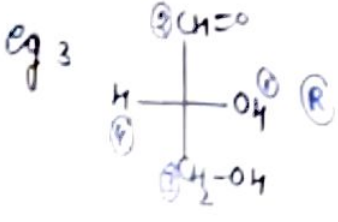
eg 1



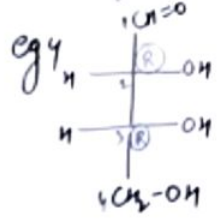
eg 2



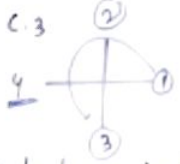
eg 3



eg 4

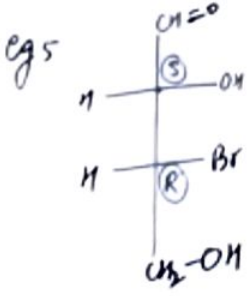




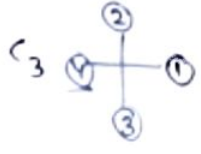


(2R, 3R) 2,3,4-Trihydroxy butanal

eg 5







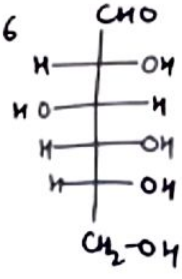
(2S, 3R).....

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

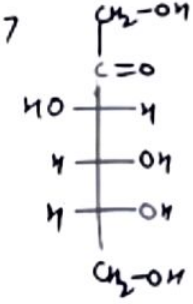
Examples of R-S Nomenclature:-

eg 6

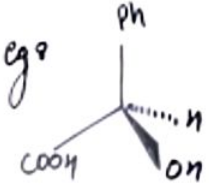


D. Glucose

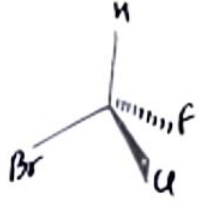
eg 7



eg 8



eg 9

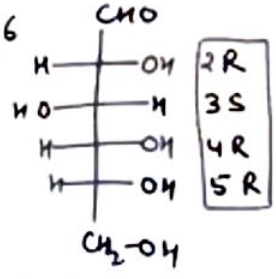


OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

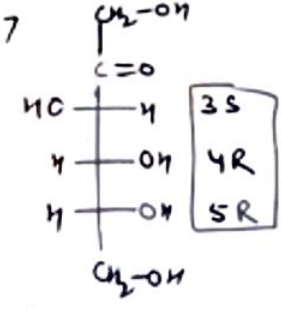
Examples of R.S Nomenclature:-

eg 6



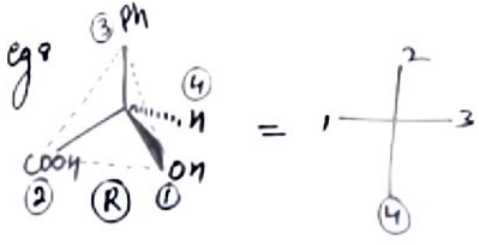
D. Glucose

eg 7

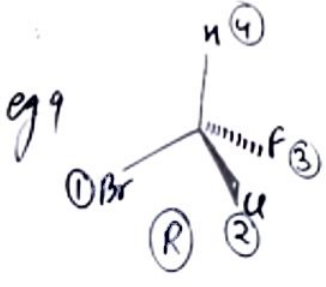


D. Fructose

eg 8



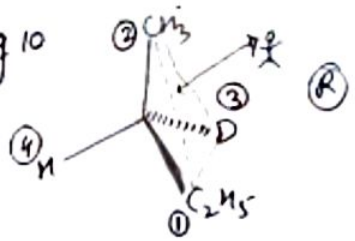
eg 9



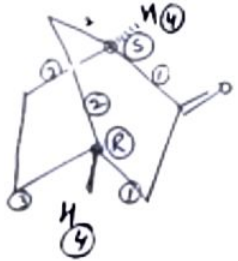
OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

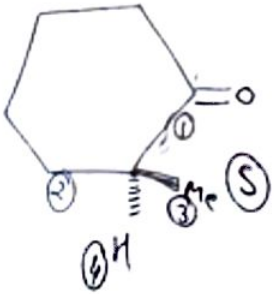
eg 10



eg 12



eg 11



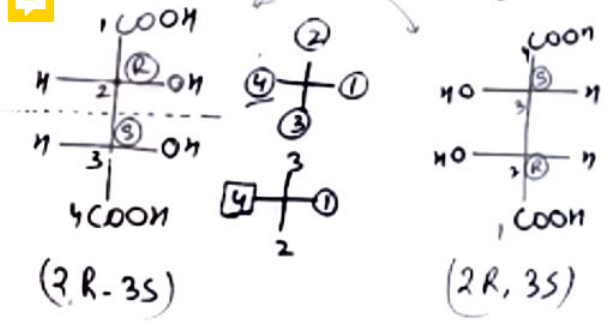


(1S, 4R) Bicyclo (2,2,1) heptan-2-one

OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

Eg 13



R > S

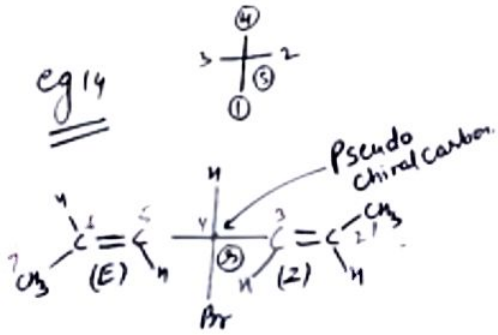
R S

(2R-3S) (2R, 3S)

Cis > Trans

Z > E

Eg 14

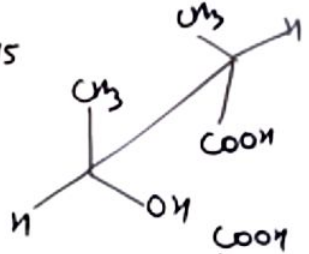


(2Z, 4S, 5E) 4-Bromohepta-2,5-diene

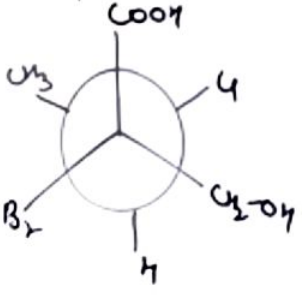
OPTICAL ISOMERISM

(XI) Configurational Nomenclature of Chiral Carbons:-

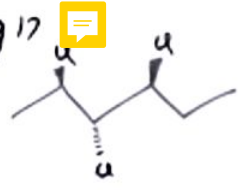
eg 15



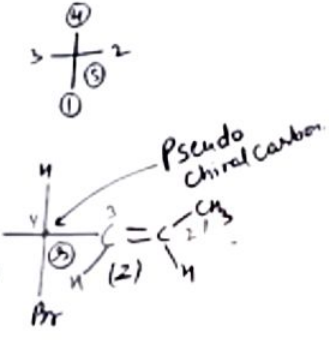
eg 16



eg 17



eg 14

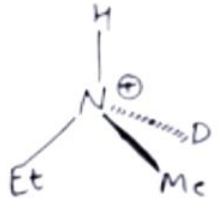


(2Z, 4S, 5E) 4-Bromohepta-2,5-diene

OPTICAL ISOMERISM

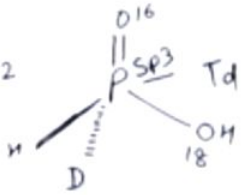
(XII) Examples of Chiral centers (other than Carbon)

eg 1



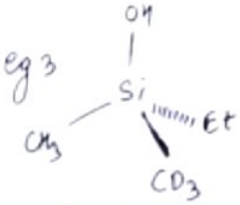
POS = X
COS = X
(T_d) OAC

eg 2



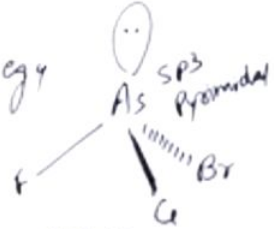
POS = X
COS = X
(OAC)

eg 3



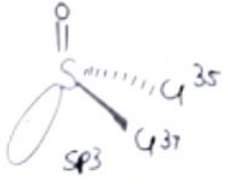
POS = X
COS = X
(OAC)

eg 4



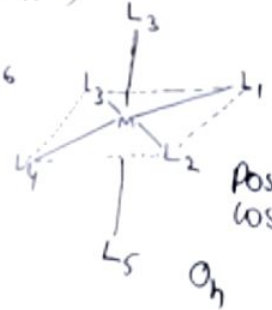
POS = X
COS = X
(OAC)

eg 5



POS = X
COS = X
} OAC.

eg 6



POS = X
COS = X
} OAC

O_h

OPTICAL ISOMERISM

(XII) Examples of Chiral centers (other than Carbon)

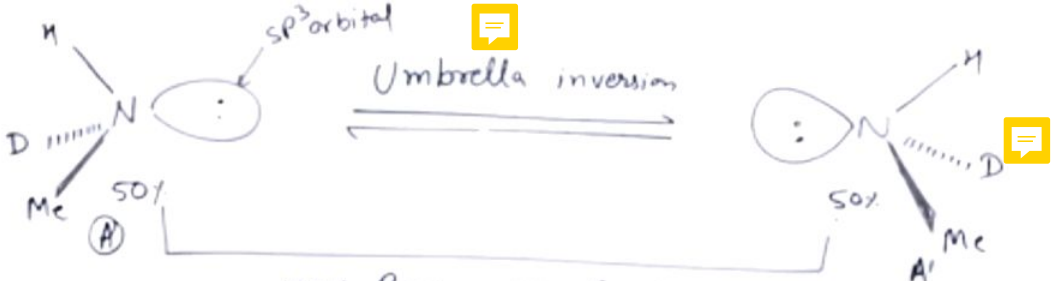
Umbrella inversion:-

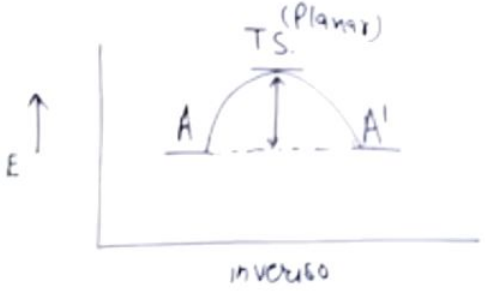


When N, C and O have three different group with one lone pair then this types of structures are optically inactive (at room temp.) It is due to umbrella inversion, By is inversion above species form racemic mix.

OPTICAL ISOMERISM

(XII) Examples of Chiral centers (other than Carbon)





NH_3
2×10^{11} /ps

OPTICAL ISOMERISM

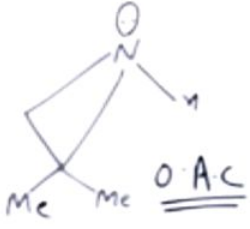
(XII) Examples of Chiral centers (other than Carbon)

⇒ Umbrella inversion not possible for 3rd and higher period elements at room temp

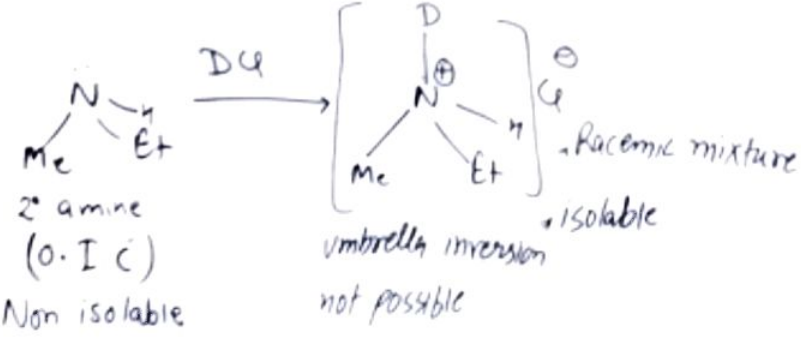
- When asymmetric nitrogen is present in small ring (upto 5 membered ring) then Umbrella inversion not possible due to high angle strain in intermediate (T.S). Such type of compound are optically active

OPTICAL ISOMERISM

(XII) Examples of chiral centers (other than Carbon)



eg



OPTICAL ISOMERISM



mples Of Optical active Compounds Without Chiral Atom.

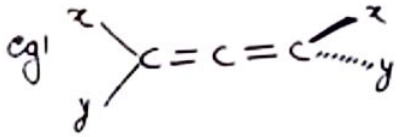
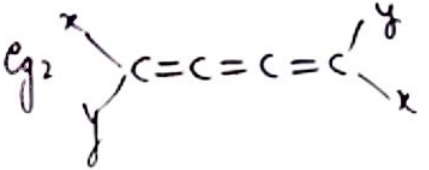
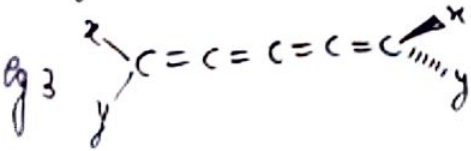
mulenes = Allenes

$CH_2=C=CH_2$

OPTICAL ISOMERISM

(XIII) Examples Of Optical active Compounds Without Chiral Atom.

[A] Cumulenes = Allenes

	Terminal x y groups	optical isomerism	G.I
eg1  $x \neq y$	$\perp$, POS = X, COS = X	✓	X
eg2  $x \neq y$	in same plane, POS = ✓	X	✓
eg3  $x \neq y$	$\perp$, POS = X, COS = X	✓	X

OPTICAL ISOMERISM

(XIII) Examples Of Optical active Compounds Without Chiral Atom.

[A] Cumulenes = Allenes

Condition for optical activity in cumulenes.

i) Terminal $x \neq y$

ii) even no. of $C=C$ double bond.

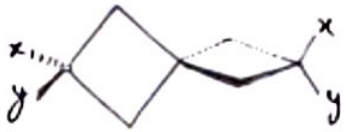


OPTICAL ISOMERISM

(XIII) Examples Of Optical active Compounds Without Chiral Atom.

(B) Spiro compounds

eg1



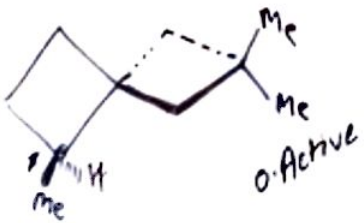
$x \neq y$ | POS = ✗ COS = ✗ | optical isomerism = ✓ | G.I = ✗

eg2



$x \neq y$ | POS = ✓ | O.Iso = ✗ | G.I = ✓

eg



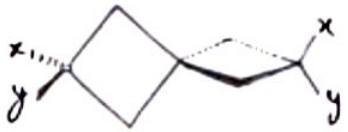
O.Active

OPTICAL ISOMERISM

(XIII) Examples Of Optical active Compounds Without Chiral Atom.

(B) Spiro compounds

eg 1



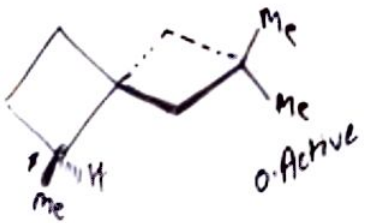
$x \neq y$ | POS = ✓ | COS = ✗ | optical isomerism = ✓ | G.I = ✗

eg 2



POS = ✓ | O.Iso = ✗ | G.I = ✓

eg 3



OPTICAL ISOMERISM

(XIII) Examples Of Optical active Compounds Without Chiral Atom.

(B) Spiro compounds

Conditions for optical activity in Spiro compounds

(i) $x \neq y$

(ii) enven no of spiro ring.

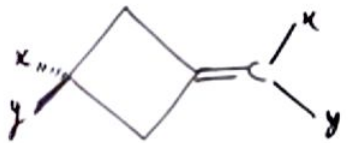
OPTICAL ISOMERISM

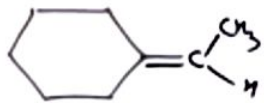
(XIII) Examples Of Optical active Compounds Without Chiral Atom.

Alkylidene cycloalkane:-

CH_3- Alkyl $CH_3-CH<$ Alkylidene

Eg 1





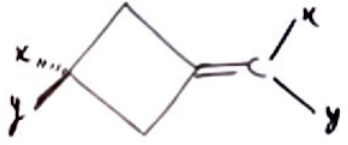
Ethylidenecyclohexane

OPTICAL ISOMERISM

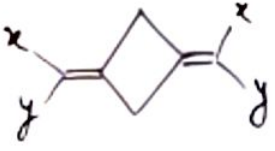
(XIII) Examples Of Optical active Compounds Without Chiral Atom.

(c) Alkylidene cycloalkane:-

CH_3- Alkyl $CH_3-CH<$ Alkylidene

eg 1 

$x \neq y$, pos = x, los = x, optical active comp.

eg 2 

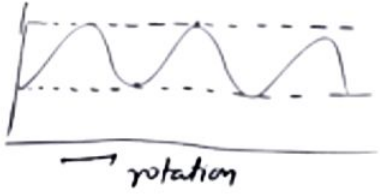
$x \neq y$ pos = v Optical inactive.

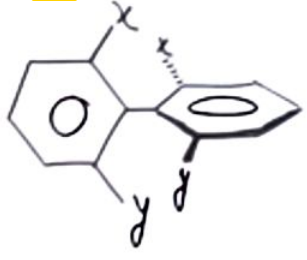
OPTICAL ISOMERISM

(XIII) Examples Of Optical active Compounds Without Chiral Atom.

(D) Ortho substituted biphenyl (Ph-Ph)

eg 



eg 2 

$\Rightarrow r_x + r_y \geq 2.9\,A^\circ$; No free rotation (Restricted rotation)

$\Rightarrow x \neq y \neq H \neq D \neq F$ etc.

* POS = X
  COS = X
  } optical active

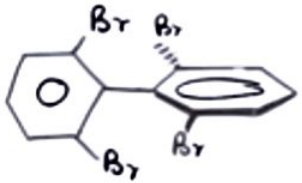
} Atropisomerism

#40|OPTICAL ISOMERISM|STEREO ISOMERISM|JEE MAIN, IT ADVANCED|BY NEERAJ SAINI|NS ...

(XIII) Examples Of Optical active Compounds Without Chiral Atom.

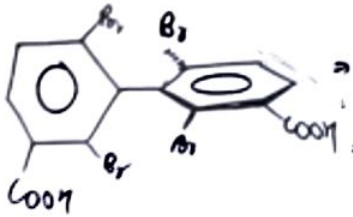
(D) Ortho substituted biphenyl (Ph-Ph)

eg 3



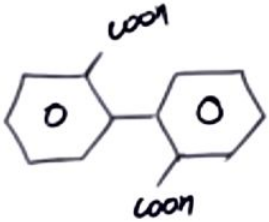
⇒ No free rotation.
⇒ POS = ✓
⇒ Optical inactive

eg 4



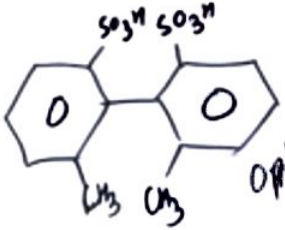
⇒ POS = ✗
COS = ✗
= Optical active

eg 5



Optical inactive

eg 6



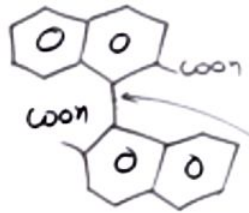
Optical active

OPTICAL ISOMERISM

(XIII) Examples Of Optical active Compounds Without Chiral Atom.

(D) Ortho substituted biphenyl (Ph-ph)

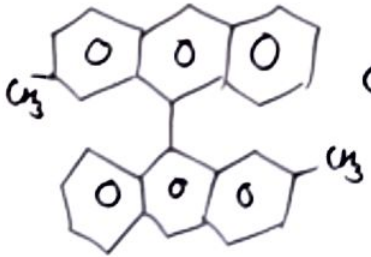
eg 7



optical active

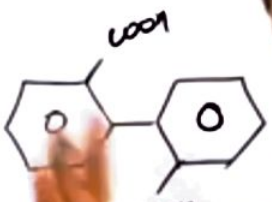
Restricted rotation.

eg 8

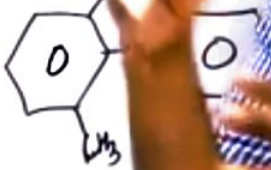


optical active comp.

eg 5



eg 6

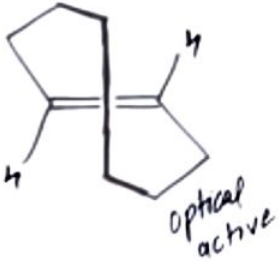


OPTICAL ISOMERISM

(XIII) Examples Of Optical active Compounds Without Chiral Atom.

(E) Trans cycloalkene :-

eg! Trans cyclo octene



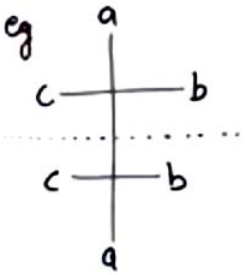
- other example



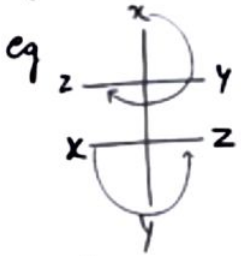
OPTICAL ISOMERISM

Short trick for identification of meso compounds

For Fischer formula

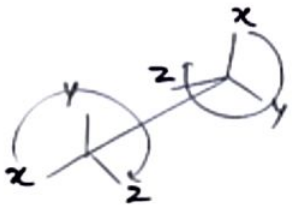


meso

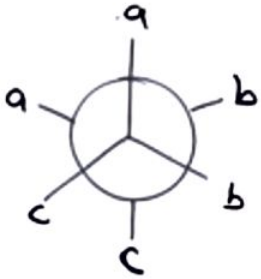


Meso

For Sawhorse and Newmann



Meso

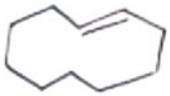


Meso

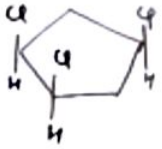
OPTICAL ISOMERISM

Q Identify Optical active, Optical inactive and meso compounds

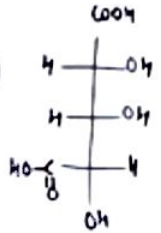
(1)



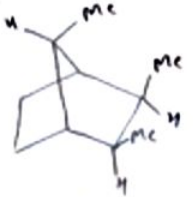
(2)



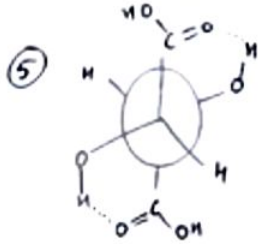
(3)



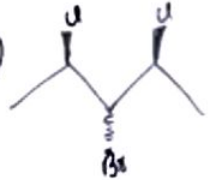
(4)



(5)



(6)



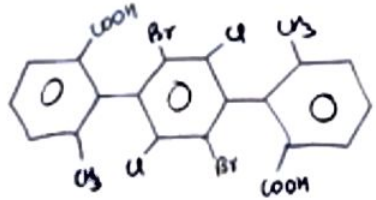
(7)



(8)

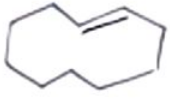


(9)

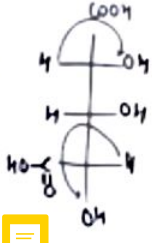


OPTICAL ISOMERISM

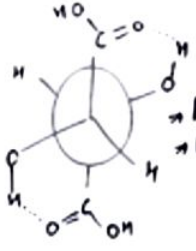
Q Identify Optical active, Optical inactive and meso compounds

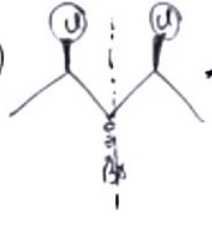
(1)  → Optical active

(2)  → Optical inactive → Meso

(3)  → Optical inactive → Meso

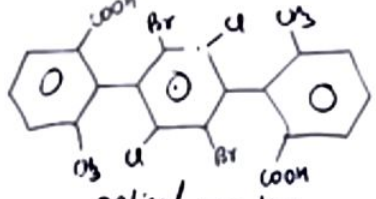
(4)  → optical inactive → Meso

(5)  → Meso → Op. inactive

(6)  → meso → Op. inactive

(7)  optical active → POS=X COS=X

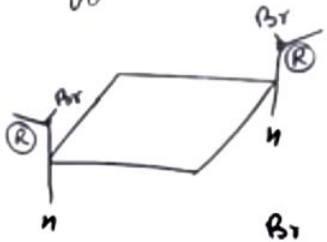
(8)  Optical active

(9)  optical inactive COS=✓

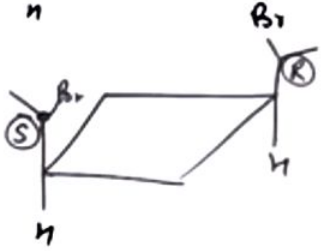
OPTICAL ISOMERISM

Q Identify Optical active, Optical inactive and meso compounds

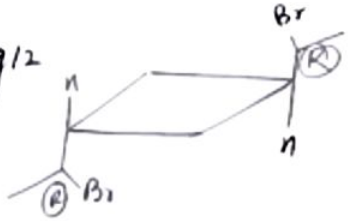
eg 10



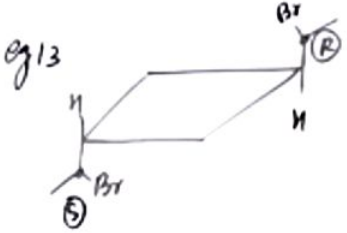
eg 11



eg 12



eg 13

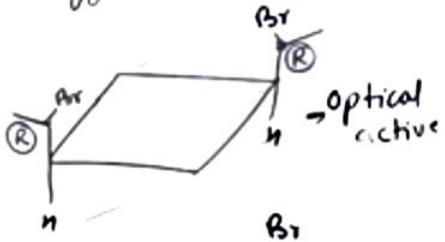




OPTICAL ISOMERISM

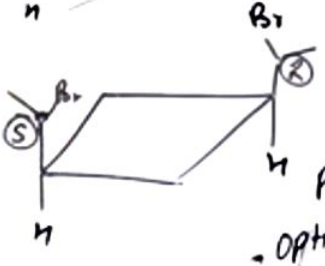
Q Identify Optical active, Optical inactive and Meso compounds

eg 10



→ optical active

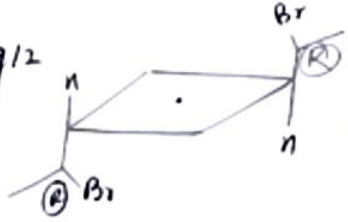
eg 11



pos = 1

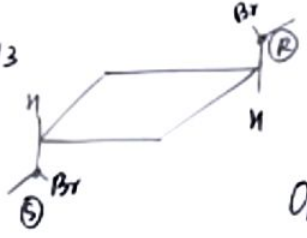
- Optical inactive
- meso

eg 12



pos = ✗
cos = ✗
Optical active

eg 13



pos = ✗
cos = ✓
Optical inactive comp

OPTICAL ISOMERISM

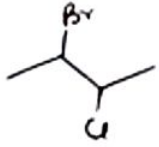
(XIV) Calculation Of Number Of Stereoisomers

	n = no. of Stereo centers	no. of optical active isomers	no. of optical inactive isomers
①	For dissimilar ends	2^n	zero
②	For similar ends n = even no.	2^{n-1}	$2^{\frac{n}{2}-1}$
③	For similar ends n = odd no. with Pseudo stereo center	$2^{n-1} - 2^{\frac{n-1}{2}}$	$2^{\frac{n-1}{2}}$
④	For similar ends n = odd no. without Pseudo stereo center.	2^{n-1}	$2^{\frac{n-1}{2}}$

OPTICAL ISOMERISM

(XIV) Calculation Of Number Of Stereoisomers

eg1



a) = Dissimilar ends

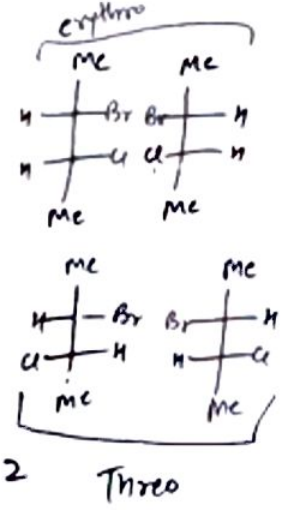
b) n = 2

c) no of op. Ac iso = $2^2 = 4$ (d,l) (dl)

d) no of op. Inactive iso. = 0

e) no of pairs of enantiomer = 2

f) no of fractions on fractional distillation of all stereoisomer (mix of) = 2

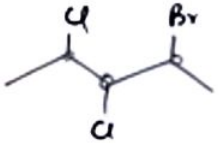


#42|OPTICAL ISOMERISM|STEREOISOMERISM|JEE MAIN, IIT ADVANCED|BYNEERAJ SAINI|NS S...

OPTICAL ISOMERISM

(IV) Calculation Of Number Of Stereoisomers

eg 2



→ dissimilar ends

→ $n = 3$

→ no. of optical active isomers $= 2^n = 2^3 = 8$

→ no. of optical inactive iso $= 0$

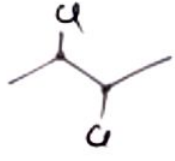
→ Pair of enantiomers $= 4$

→ no. of fractions on fractional distillation $= 4$

OPTICAL ISOMERISM

(XIV) Calculation Of Number Of Stereoisomers

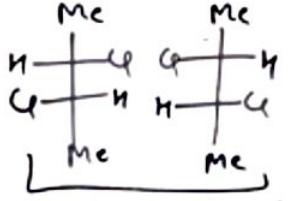
Eg 3



→ Similar ends

⇒ $n = 2$

→ no of optical active isomers = $2^{n-1} = 2$ (d, l) ✓



Threo Enanty.

⇒ no. of optical inactive iso. = $2^{\frac{n}{2}-1} = 2^0 = 1$ (meso) ✓

⇒ Total Stereo isomers = 3

⇒ no of pair of enantiomer = 1

⇒ no of fractions on fractional distillation = 2 (d l) (meso)



OPTICAL ISOMERISM

(XIV) Calculation Of Number Of Stereoisomers

eg 4

eg 5

eg 6

Same as eg no. 3

eg 5

optical active = (2) ← Trans.

optical inactive = 1 meso ← cis

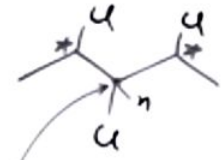
Optical active

(cis) Meso

OPTICAL ISOMERISM

(XIV) Calculation Of Number Of Stereoisomers

Eg 1



Pseudo Chiral Carbon

⇒ Similar ends.

⇒ n = 3

⇒ no. of optical active iso. = $2^{n-1} - 2^{\frac{n-1}{2}} = 4 - 2 = 2$ (d, l)

⇒ no. of optical inactive iso. = $2^{\frac{n-1}{2}} = 2$ (meso) (meso)

= Total no of Stereoiso. = 4

⇒ Pairs of enantiomers = 1

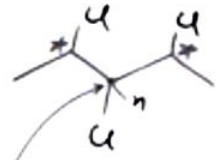
⇒ no. of fractions on fra. dis. = 3

NEERAJ SAINI (N.S. SIR)

OPTICAL ISOMERISM

(XIV) Calculation Of Number Of Stereoisomers

Eg 1



Pseudo Chiral Carbon

⇒ Similar ends.

⇒ n = 3

⇒ no. of optical active iso. = $2^{n-1} - 2^{\frac{n-1}{2}} = 4 - 2 = 2$ (d, l)

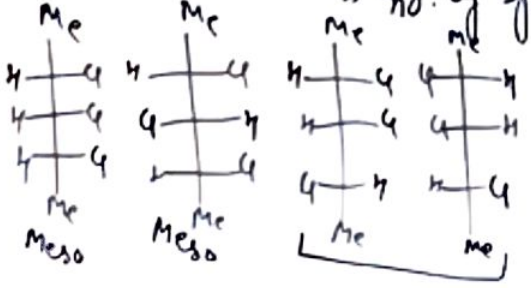
⇒ no. of optical inactive iso. = $2^{\frac{n-1}{2}} = 2$ (meso) (meso)

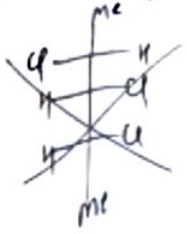
= Total no of Stereoiso. = 4

⇒ Pairs of enantiomers = 1

⇒ no. of fractions on fra. dis. = 3

① Me

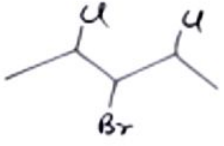




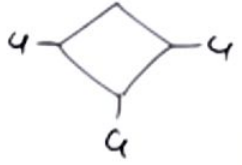
OPTICAL ISOMERISM

(XIV) Calculation Of Number Of Stereoisomers

eg 8



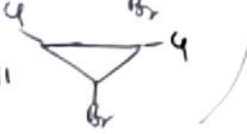
eg 9



eg 10



eg 11



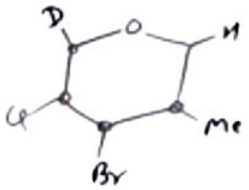
Same as eg 7

eg 12 $CH_3-\overset{*}{C}H=CH-\overset{*}{C}H-CH_3$ (n = 2)
$\quad\quad\quad

OPTICAL ISOMERISM

(XIV) Calculation Of Number Of Stereoisomers

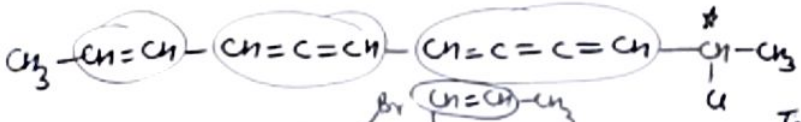
eg 13



$n = 4$

Optical active ison $= 2^4 = 16$

eg 14

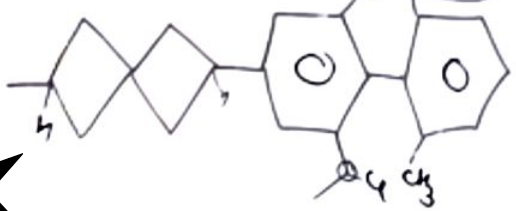


$n = 4$

Total stereoisomer $= 2^4 = 16$

no. of optical active $= 16$

eg 15



$n = 4$

Total stereo isomer $= 2^4 = 16$



OPTICAL ISOMERISM

(XIV) Calculation Of Number Of Stereoisomers

eg 16 $CH_3-CH=CH-\underset{\underset{Cl}{|}}{CH}-CH=CH-CH_3$ (Pseudo Chiral)

→ Similar ends

→ $n = 3$

→ Optical active isomer $= 2^{n-1} - 2^{\frac{n-1}{2}} = 2$

→ " inactive " " $= 2^{\frac{n-1}{2}} = 2$

→ Total stereoisomer $= \frac{}{4}$

① Cis —— Cis]
② Trans —— Trans] optical inactive

③ Cis — R — Trans]
④ Cis — S — Trans.] optical active.

OPTICAL ISOMERISM

(XIV) Calculation Of Number Of Stereoisomers

eg 17 $CH_3-\underset{\underset{Br}{|}}{CH}-CH=CH-\underset{\underset{Br}{|}}{CH}-CH_3$

- Similar ends
- $n=3$
- Optical active isomer $= 2^{n-1} = 2^{3-1} = 4$
- " inactive " " $= 2^{\frac{n-1}{2}} = 2$
- Total stereoisomers = 6

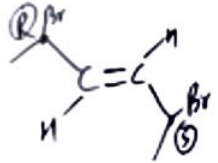
① R — Cis — R
② S — Cis — S
} enantiom.

③ R — Cis — S } meso
Pos=V



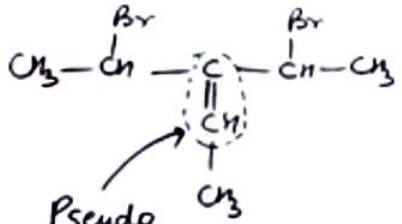
④ R — Trans — R
⑤ S — Trans — S
} enantiom.

⑥ R — Trans — S } meso
Cos=V



(XIV) Calculation Of Number Of Stereoisomers

eg 18



- Similar ends.
- ⇒ n = 3
- ⇒ Optical active = $2^{n-1} - 2^{\frac{n-1}{2}} = 2$
- ⇒ Optical inactive = $2^{\frac{n-1}{2}} = 2$

eg 19

① R——R] optical active
② S——S]

③ R—E—S] optical inactive
④ R—Z—S]

OPTICAL ISOMERISM

Q Identify all possible comp.

① C_4H_9Cl DU=0 ⇒ 5

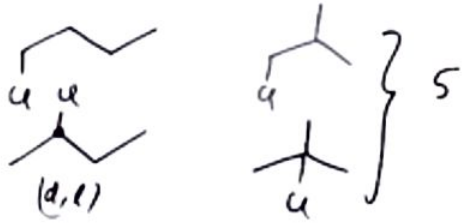
② $C_5H_{11}Cl$ DU=0 ⇒ 11

③ Dichloro cyclo butane

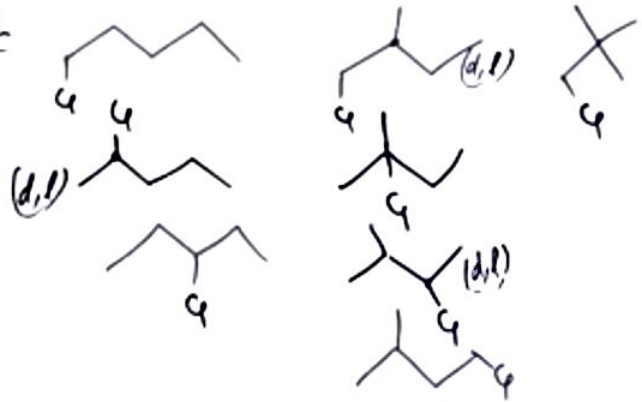
④ Trichloro cyclobutane

⑤ Dibromochloro cyclobutane

Sol. 1



Sol. 2



OPTICAL ISOMERISM

Q. Identify all possible comp.

D.U=0 ⇒ 5

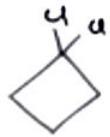
D.U=0 ⇒ 11

chloro cyclo butane ⇒ 6

chloro cyclobutane ⇒ 7

dichloro cyclobutane

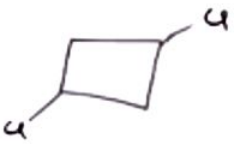
Sol 3



(1)

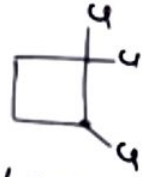


(d, l, meso)

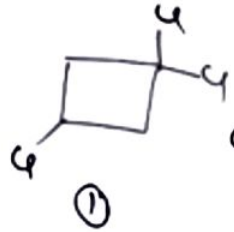


(Cis + Trans)

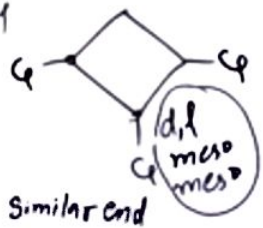
Sol-4



(d, l)



(1)



Similar end

n = 3

O. Active = $2^{n-1} - 2^{\frac{n-1}{2}} = 2$

O. Inactive = $2^{\frac{n-1}{2}} = 2$

OPTICAL ISOMERISM



Q Identify all possible comp.

① $C_4H_8Cl_2$ D.U=0 ⇒ 5

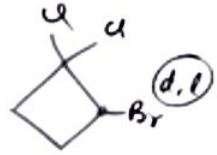
② $C_5H_{11}Cl$ D.U=0 ⇒ 11

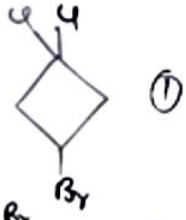
③ Dichloro cyclobutane ⇒ 6

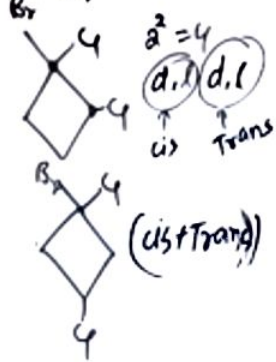
④ Trichloro cyclobutane ⇒ 7

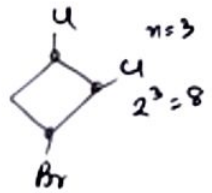
⑤ Dibromochloro cyclobutane ⇒ 21

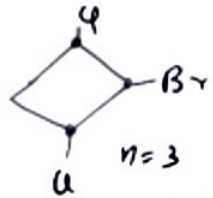
Sol.5











O.Active = 2

O.Inctive = 2

4

OPTICAL ISOMERISM

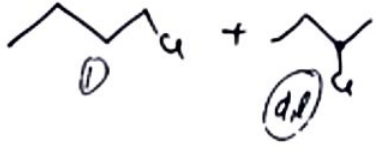
Q Identify all possible monochloro products.

Hint

$-\overset{|}{\underset{|}{C}}-H \xrightarrow[-HCl]{Cl_2 + h\nu} -\overset{|}{\underset{|}{C}}-Cl$

(C: sp^3)

monochloro product

①  $\xrightarrow[\text{monochlorination}]{Cl_2 + h\nu}$ 

(D) + (d,l)

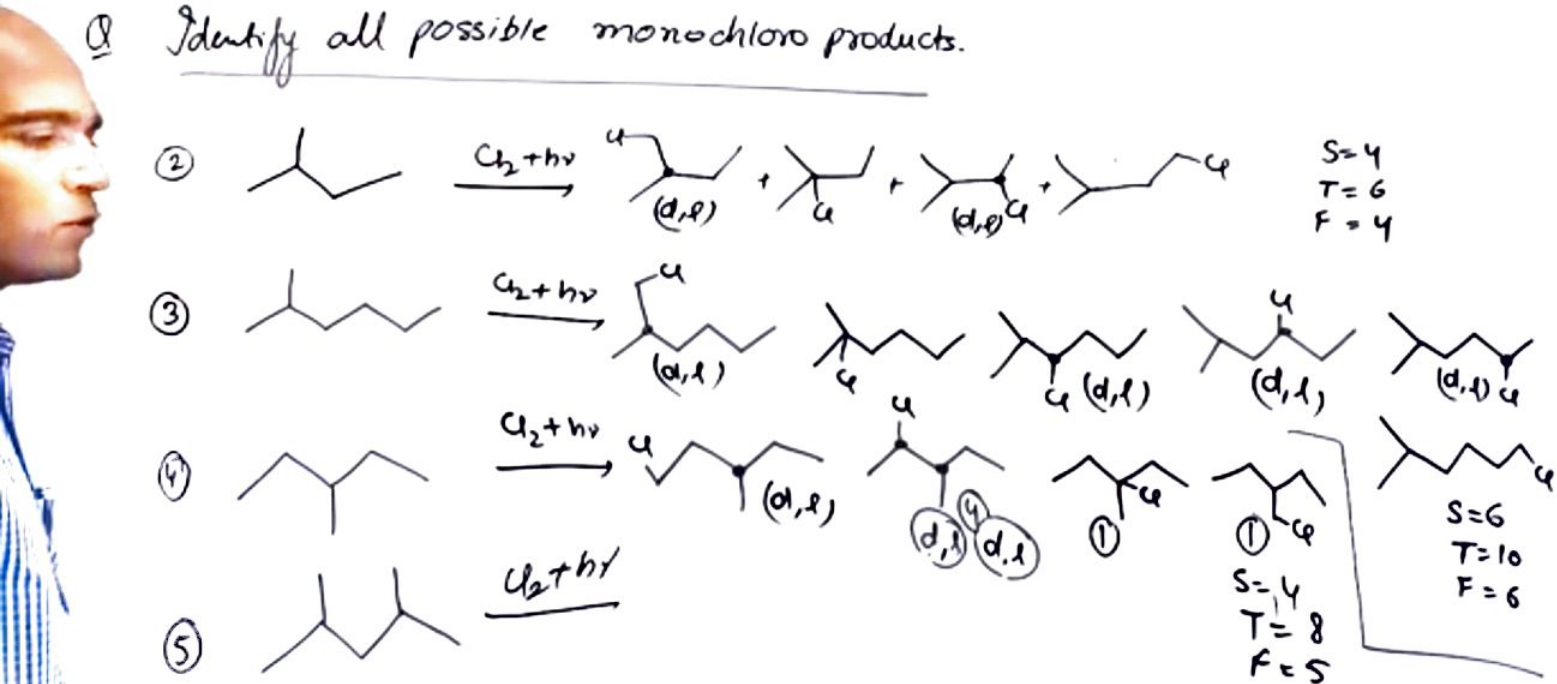
S = Structural iso. products = 2

T = Total Product = 3

F = Total no. of fractions on fractional distillation of mix. of product = 2

OPTICAL ISOMERISM

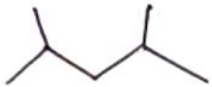
Q Identify all possible monochloro products.

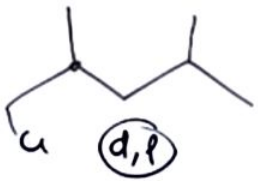


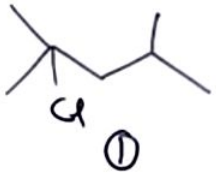
#44|OPTICAL ISOMERISMISTEREOISOMERISM|JEE MAIN, IT ADVANCED|BYNEERAJ SAINI|NS S...

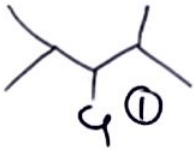
OPTICAL ISOMERISM

Q Identify all possible monochloro products.

(5)  $Cl_2 + h\nu$ ⟶





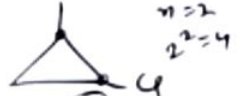


S = 3
T = 4
F = 3

OPTICAL ISOMERISM

Q Identify all possible monochloro products.

(6)  $Cl_2 + h\nu$ →  (I)  (II)



(4) d,l ← Trans
d,l ← cis

S = 3
T = 6
F = 4

(7)  $Cl_2 + h\nu$ →







O.Active = 2 d,l
O.Inactive = 2 meso meso

S = 3
T = 12
F = 7

(8)  $Cl_2 + h\nu$ →



O.Active = 2
O.Inactive = 2

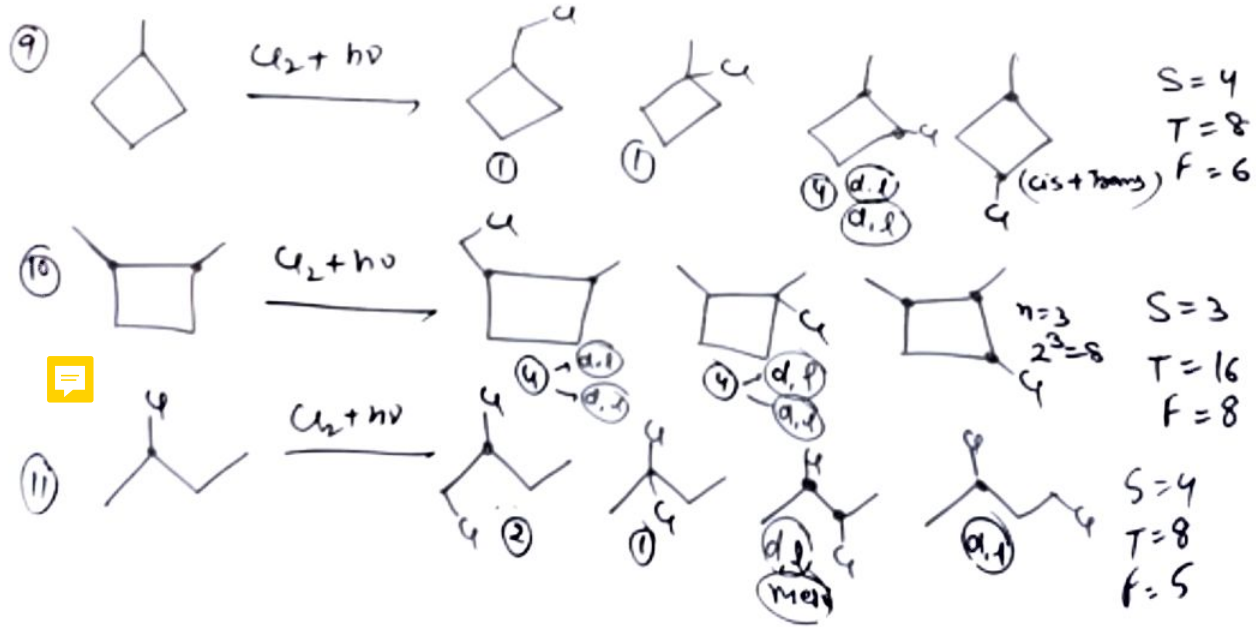


O.Active = 2 ✓
O. Inactive = 2

S = 2
T = 8
F = 6

OPTICAL ISOMERISM

Q Identify all possible monochloro products.



OPTICAL ISOMERISM

Q Identify all possible monochloro products.

② 